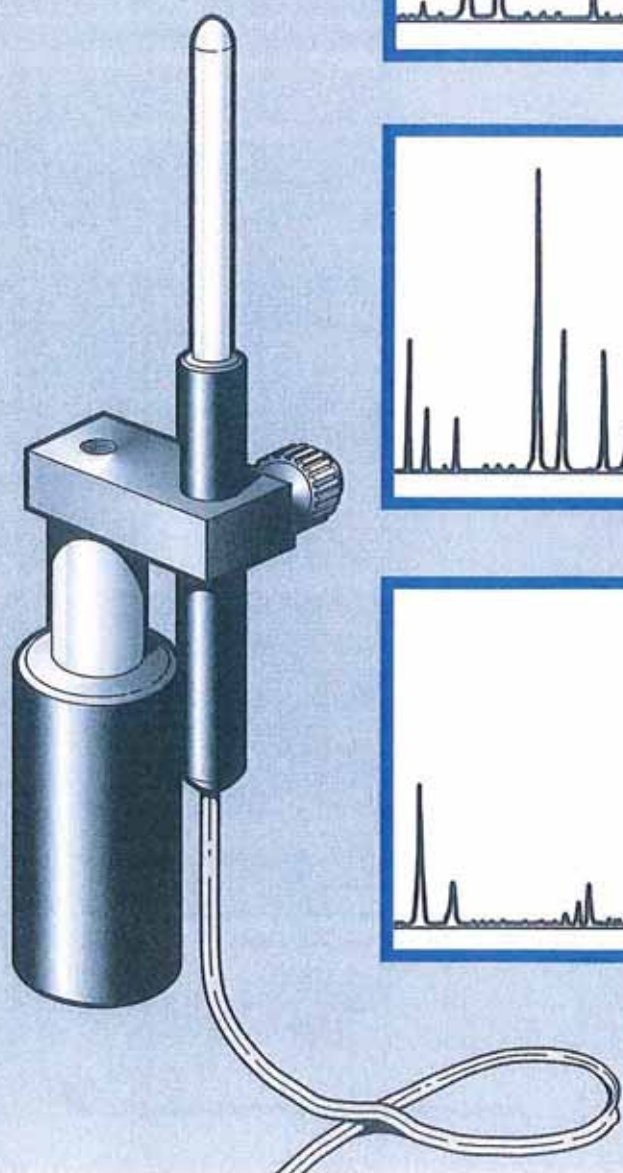
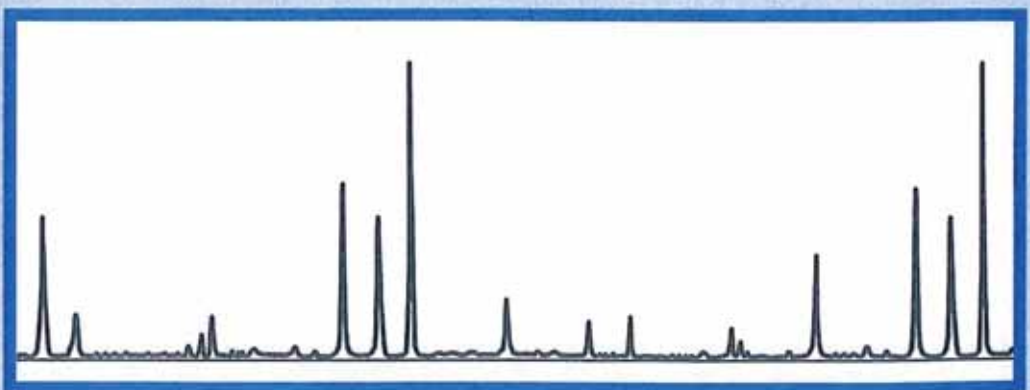
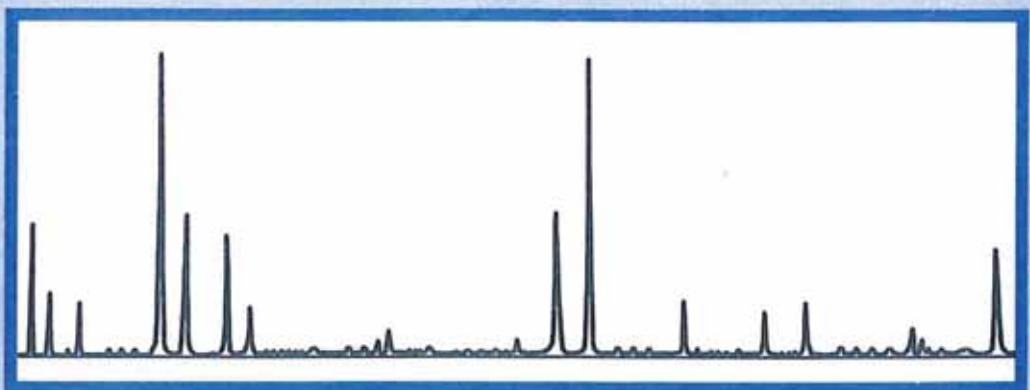
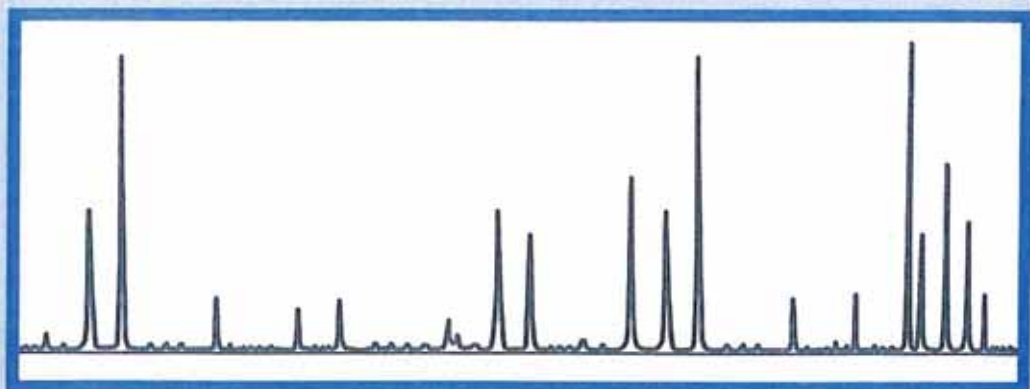


Typical Spectra of
DRIEL
INSTRUMENTS
Spectral Calibration Lamps



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Our Spectral Calibration Lamps are compact and convenient to use. Both AC and DC Power Supplies are offered to run the lamps at various operating currents, and a rod mounted holder is available to support the lamps. Several lamp shields are also offered to aperture the lamps or for spectral filtering and conversion.

These lamps and their accessories are described in detail in Volume II, pages 27 to 30. This booklet contains their spectral output. We offer the following types:

- Argon
- Krypton
- Neon
- Xenon
- Mercury (Argon)
- Mercury-Neon



6035 Hg(Ar) Lamp in 63670 Lamp Holder, with 6047 Power Supply.

SAFETY

Although these are not considered powerful lamps, some of them have a strong ultraviolet output. You should avoid unnecessary exposure of the eyes or skin to shortwave ultraviolet. Apart from the possible mutagenic effects of ultraviolet radiation, you can suffer delayed corneal burns ("welder's eye") from working close to some of the mercury lamps for an hour or so. These burns are at the least extremely uncomfortable, and possibly worse.

You may not notice anything amiss (except perhaps watery eyes), while working with one of the ultraviolet emitting lamps. If you get too much exposure, and this has happened in our laboratories, several hours later you will have difficulty focusing and considerable eye discomfort. This can persist for many hours and in addition to the discomfort, it is extremely frightening. So, if you need the ultraviolet output, make sure you use goggles and gloves. (We carry these in Volume II.) If you do not need the ultraviolet, then we recommend you use the 6057 Glass Safety Filter on the lamps. The 6057 absorbs the ultraviolet while transmitting the visible and infrared calibration lines.

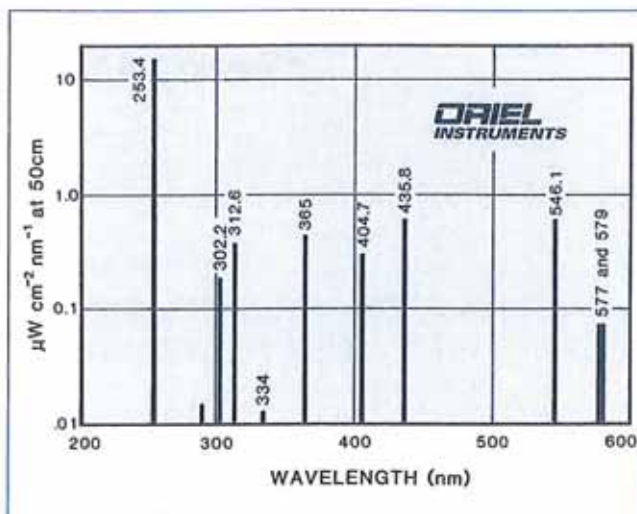


Fig. 1 Typical Irradiance at 50 cm from 6035 Hg(Ar) Lamp operated at 18 mA.

LAMP CHARACTERISTICS

The Hg (Ar) and Hg-Ne lamps contain a small amount of mercury which dominates the output spectrum, and argon or neon as a starter gas. The spectral output of the 6035 Hg(Ar) lamp is effectively that of the mercury lines. These lamps are insensitive to temperature. With the 18 mA power supplies, 90% of the output is in the 253.7 nm line; with the 30 mA power supplies, intensity shifts to the visible lines and only 30% is left in the 253.7 nm line.

The Hg-Ne lamp, model 6034, is temperature sensitive. In normal lab ambient, the output is similar to the 6035, i.e. dominated by mercury lines. To produce the neon lines, forced air cooling is required. Operate the 6034 at 18 mA only.

The rare gas calibration lamps operate at 6 or 10 mA.

Table 1, on the following page, lists the useful wavelengths for our spectral calibration lamps.

Table 1 Useful Wavelengths for Spectral Calibration Lamps

Wavelength (nm)	Lamp	Model No.	Wavelength (nm)	Lamp	Model No.
184.91	Hg*	6035	433.36	Ar	6030
194.17	Hg*	6035	433.92	Hg*	6035
226.22	Hg*	6035	434.52	Ar	6030
237.83	Hg*	6035	434.75	Hg*	6035
248.2	Hg*	6035	435.14	Kr	6031
253.65	Hg*	6035	435.84	Hg*	6035
265.2	Hg*	6035	436.26	Kr	6031
275.28	Hg*	6035	437.61	Kr	6031
280.35	Hg*	6035	440.0	Kr	6031
289.36	Hg*	6035	442.52	Kr	6031
296.73	Hg*	6035	445.39	Kr	6031
302.15	Hg*	6035	446.37	Kr	6031
312.57	Hg*	6035	450.1	Xe	6033
313.17	Hg*	6035	450.24	Kr	6031
334.15	Hg*	6035	452.47	Xe	6031
336.99	Ne	6032	458.28	Xe	6033
341.79	Ne†	6032	462.43	Xe	6033
344.77	Ne†	6032	467.12	Xe	6033
345.42	Ne	6032	469.7	Xe	6033
346.66	Ne	6032	470.44	Ne	6032
347.26	Ne†	6032	473.42	Xe	6033
349.81	Ne	6032	480.7	Xe	6033
350.12	Ne	6032	482.97	Xe	6033
351.52	Ne	6032	484.33	Xe	6033
352.05	Ne†	6032	491.6	Hg*	6035
355.43	Ar	6030	491.65	Xe	6033
359.35	Ne†	6032	492.32	Xe	6033
360.02	Ne	6032	502.83	Xe	6033
363.37	Ne	6032	503.78	Ne	6032
365.02	Hg*	6035	508.04	Ne	6032
365.48	Hg*	6035	511.65	Ne	6032
366.33	Hg*	6035	514.5	Ne	6032
366.53	Kr	6031	520.39	Ne	6032
367.96	Kr	6031	533.08	Ne†	6032
377.34	Kr	6031	534.11	Ne†	6032
394.9	Ar	6030	534.33	Ne†	6032
404.44	Ar	6030	540.06	Ne†	6032
404.66	Hg*	6035	546.07	Hg*	6035
407.78	Hg*	6035	549.09	Kr	6031
415.86	Ar	6030	550.07	Kr	6031
416.4	Ar	6030	556.22	Kr	6031
418.19	Ar	6030	557.03	Kr	6031
419.1	Ar	6030	558.04	Kr	6031
419.8	Ar	6030	564.96	Kr	6031
420.07	Ar	6030	565.67	Ne	6032
425.12	Ar	6030	567.25	Kr	6031
425.94	Ar	6030	574.83	Ne	6032
426.63	Ar	6030	576.44	Ne	6032
427.22	Ar	6030	576.96	Hg*	6035
427.4	Ar	6030	579.07	Hg*	6035
428.3	Kr	6031	580.45	Ne	6032
430.01	Ar	6030	582.02	Ne	6032
431.96	Kr	6031	583.29	Kr	6031
432.0	Ar	6030	585.25	Ne	6032

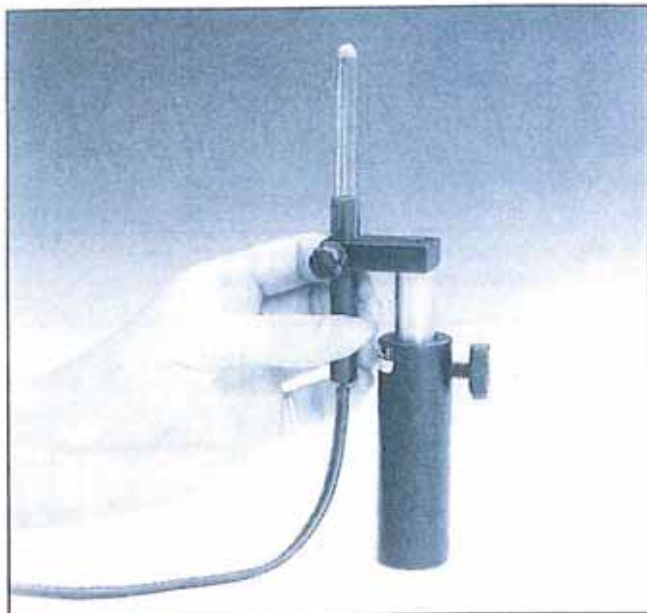
* These lines appear with the Hg(Ar) and Hg-Ne lamps.

† These lines also appear with the Hg-Ne lamp.

Wavelength (nm)	Lamp	Model No.
587.09	Kr	6031
587.99	Kr	6031
588.19	Ne†	6032
590.25	Ne	6032
594.48	Ne†	6032
597.55	Ne†	6032
598.79	Ne	6032
599.39	Kr	6031
601.22	Kr	6031
603.0	Ne	6032
605.61	Kr	6031
607.43	Ne	6032
609.62	Ne	6032
612.85	Ne	6032
614.31	Ne	6032
616.36	Ne	6032
618.22	Ne	6032
618.24	Xe	6033
621.73	Ne	6032
626.65	Ne	6032
630.48	Ne	6032
631.81	Xe	6033
633.44	Ne	6032
638.3	Ne	6032
640.23	Ne	6032
641.63	Ar	6030
642.1	Kr	6031
645.63	Kr	6031
646.97	Xe	6033
650.65	Ne	6032
653.29	Ne	6032
659.9	Ne	6032
665.21	Ne	6032
666.89	Xe	6033
667.73	Ar	6030
667.83	Ne†	6032
669.92	Kr	6031
671.7	Ne	6032
672.8	Xe	6033
675.28	Ar	6030
681.31	Kr	6031
682.73	Xe	6033
687.13	Ar	6030
688.22	Xe	6033
690.47	Kr	6031
692.95	Ne	6032
693.77	Ar	6030
696.54	Ar	6030
702.41	Ne	6032
703.03	Ar	6030
703.24	Ne	6032
705.3	Ne	6032
705.91	Ne	6032
706.72	Ar	6030
711.96	Xe	6033
714.70	Ar	6030
717.39	Ne	6032

Wavelength (nm)	Lamp	Model No.
722.41	Kr	6031
724.52	Ne†	6032
727.29	Ar	6030
728.53	Xe	6033
728.98	Kr	6031
737.21	Ar	6030
738.4	Ar	6030
742.55	Kr	6031
743.89	Ne†	6032
747.24	Ne	6032
748.61	Kr	6031
748.89	Ne†	6032
750.39	Ar	6030
751.46	Ar	6030
753.58	Ne	6032
754.41	Ne	6032
758.47	Xe	6033
758.74	Kr	6031
760.15	Kr	6031
763.51	Ar	6030
764.2	Xe	6033
768.52	Kr	6031
769.45	Kr	6031
772.38	Ar	6030
774.68	Kr	6031
785.48	Kr	6031
791.34	Kr	6031
794.32	Ne	6032
794.82	Ar	6030
800.62	Ar	6030
801.48	Ar	6030
805.95	Kr	6031
808.25	Ne	6032
810.37	Ar	6030
810.44	Kr	6031
811.29	Kr	6031
811.53	Ar	6030
813.64	Ne	6032
819.01	Kr	6031
823.16	Xe	6033
826.32	Kr	6031
826.45	Ar	6030
828.01	Xe	6033
828.11	Kr	6031
829.81	Kr	6031
830.03	Ne	6032
834.68	Xe	6033
837.76	Ne	6032
840.82	Ar	6030
840.92	Xe	6033
841.84	Ne	6032
842.46	Ar	6030
849.54	Ne	6032
850.89	Kr	6031
877.67	Kr	6031
892.87	Kr	6031

† These lines also appear with the Hg-Ne lamp.

LAMP HOLDER

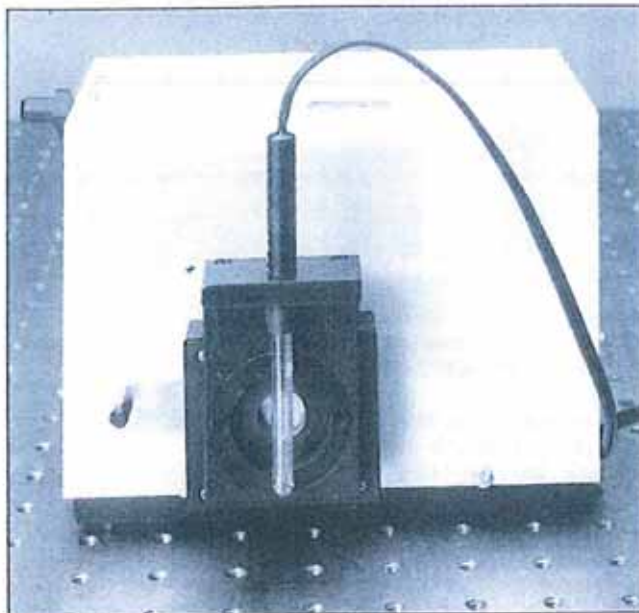
6035 Spectral Calibration Lamp being inserted in 63670 Lamp Holder, on rod and rod holder.

The 63670 is a convenient and inexpensive rod mount that holds any of our spectral calibration lamps in a vertical position. A 1.75 inch (44.5 mm) long optical rod is included. See Volume II, page 29 for ordering information for the 63670.

SPECTRAL CALIBRATION LAMP SETS

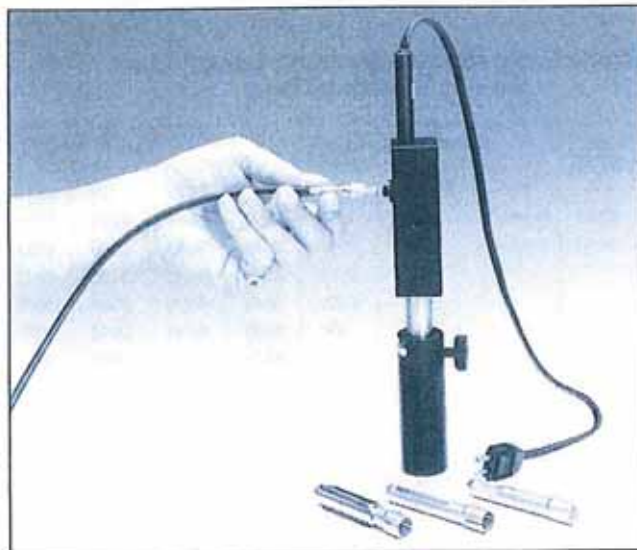
These sets include the following:

- 6030 Argon Lamp
- 6031 Krypton Lamp
- 6032 Neon Lamp
- 6033 Xenon Lamp
- 6035 Mercury (Argon) Lamp
- 6040 Shield
- 6043 115 V Power Supply or 6044 220 V Power Supply
- 6045 115 V Power Supply or 6046 220 V Power Supply
- Carrying Case and Booklet of Spectra

**LAMP HOLDER TO CALIBRATE ORIEL
MONOCHROMATORS**

77251 Calibration Lamp Holder with lamp mounted to 77250 1/8 m Monochromator.

The 77251 holds a spectral calibration lamp in front of the entrance slit of our 1/4 m or 1/8 m monochromators to calibrate the monochromator. See Volume II, page 201.

**LAMP SHIELDS/FILTERS AND FIBER OPTIC
ADAPTER**

6058 Fiber Optic Adapter on rod and rod holder with Spectral Calibration Lamps and various Lamp Shields.

Lamp shields to aperture the radiating region and filters to absorb the ultraviolet lines while transmitting the visible are offered for these lamps. The 6058 Fiber Optic Adapter lets you collect and carry a portion of the lamp output via an SMA terminated single fiber. See Volume II, page 29 for information on these accessories.

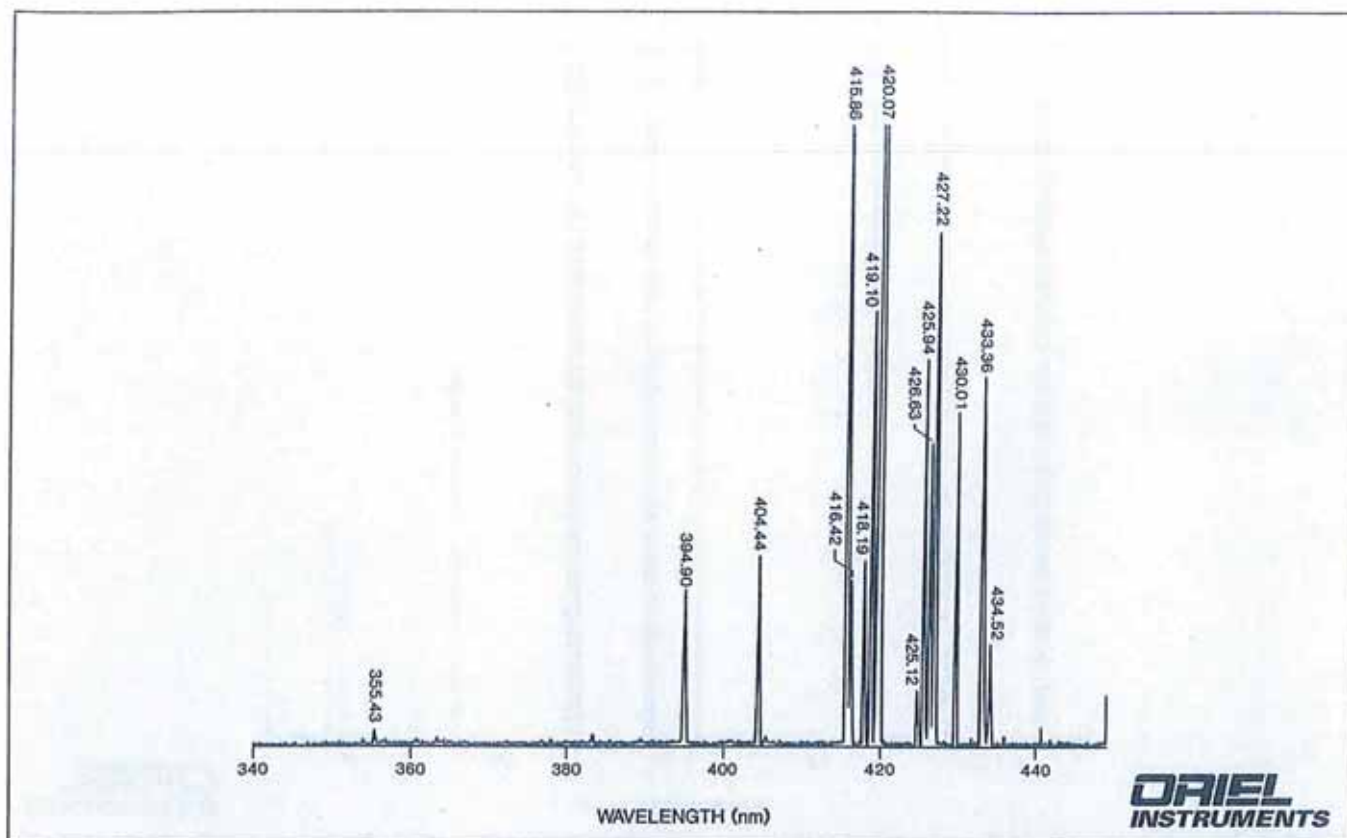


Fig. 1 Typical line output of 6030 Argon Lamp. Relative intensities vary with operating conditions.

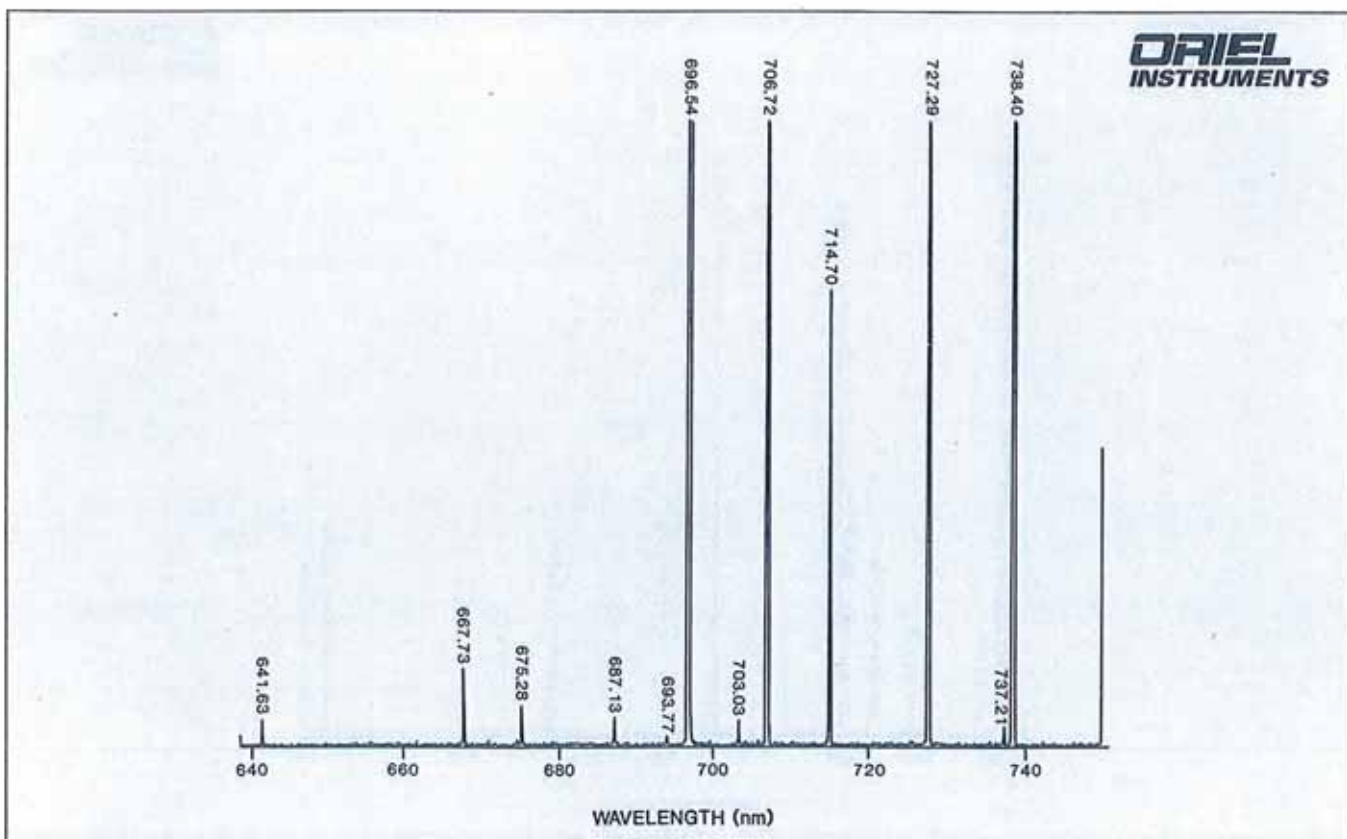


Fig. 2 Typical line output of 6030 Argon Lamp. Relative intensities vary with operating conditions.

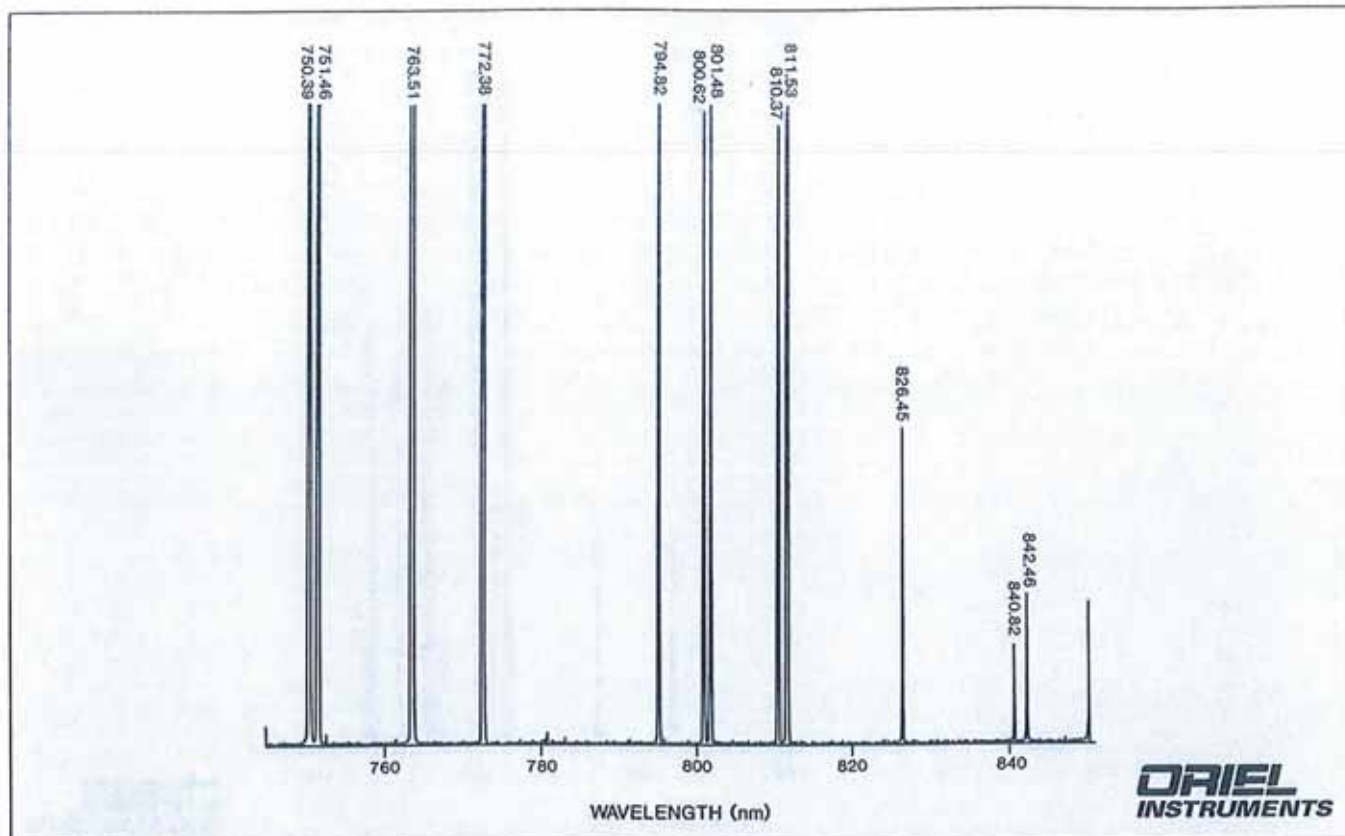


Fig. 3 Typical line output of 6030 Argon Lamp. Relative intensities vary with operating conditions.

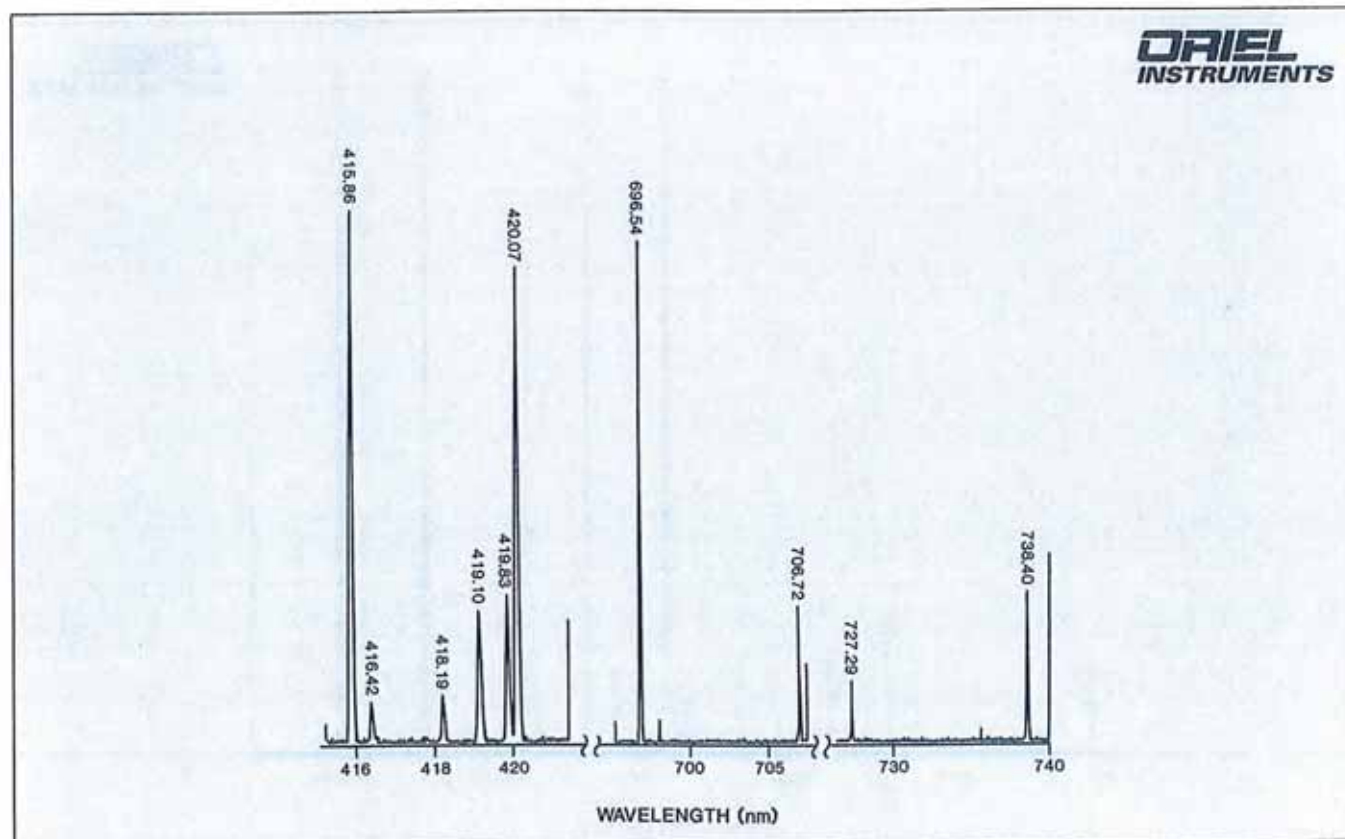


Fig. 4 Typical line output of 6030 Argon Lamp. Relative intensities vary with operating conditions.

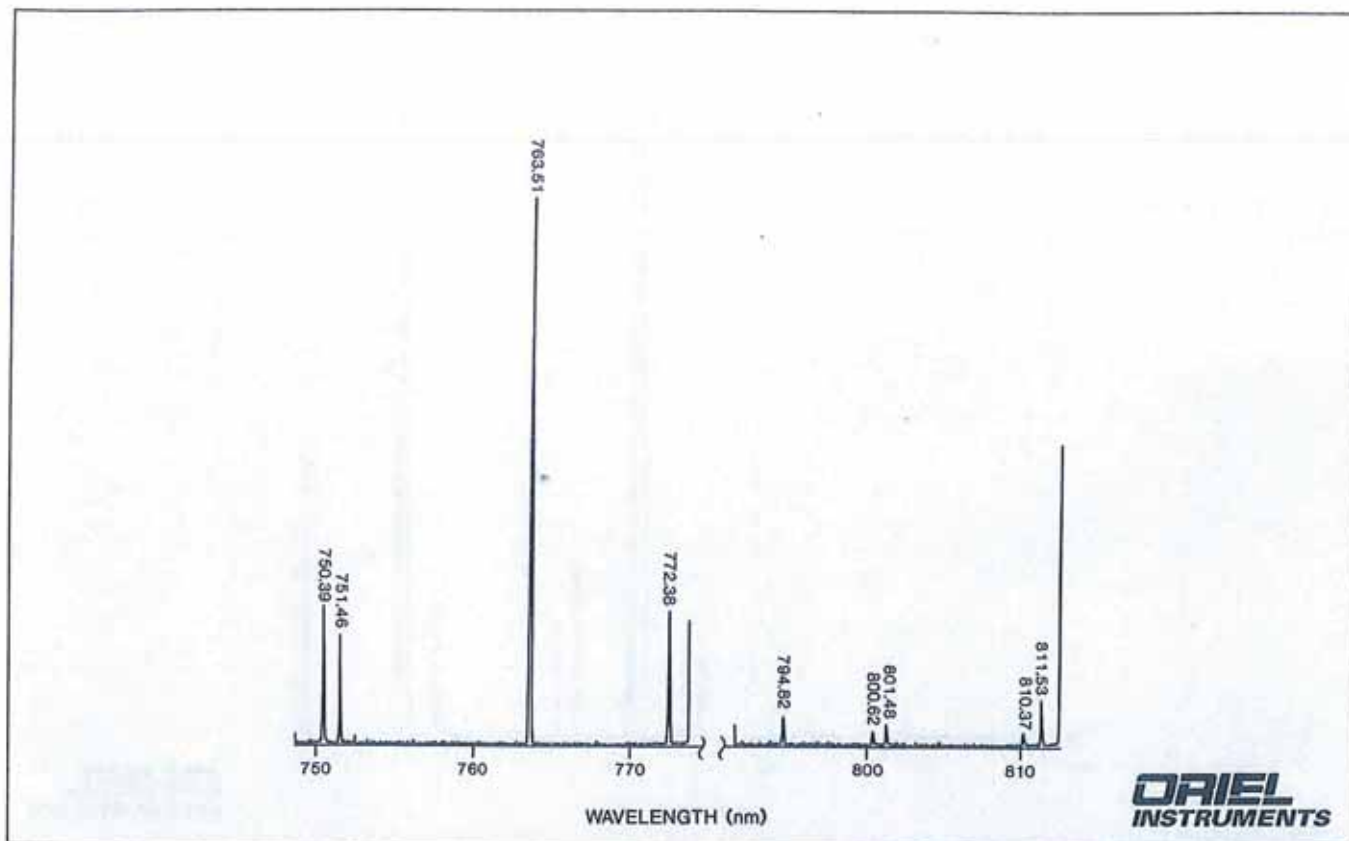


Fig. 5 Typical line output of 6030 Argon Lamp. Relative intensities vary with operating conditions.

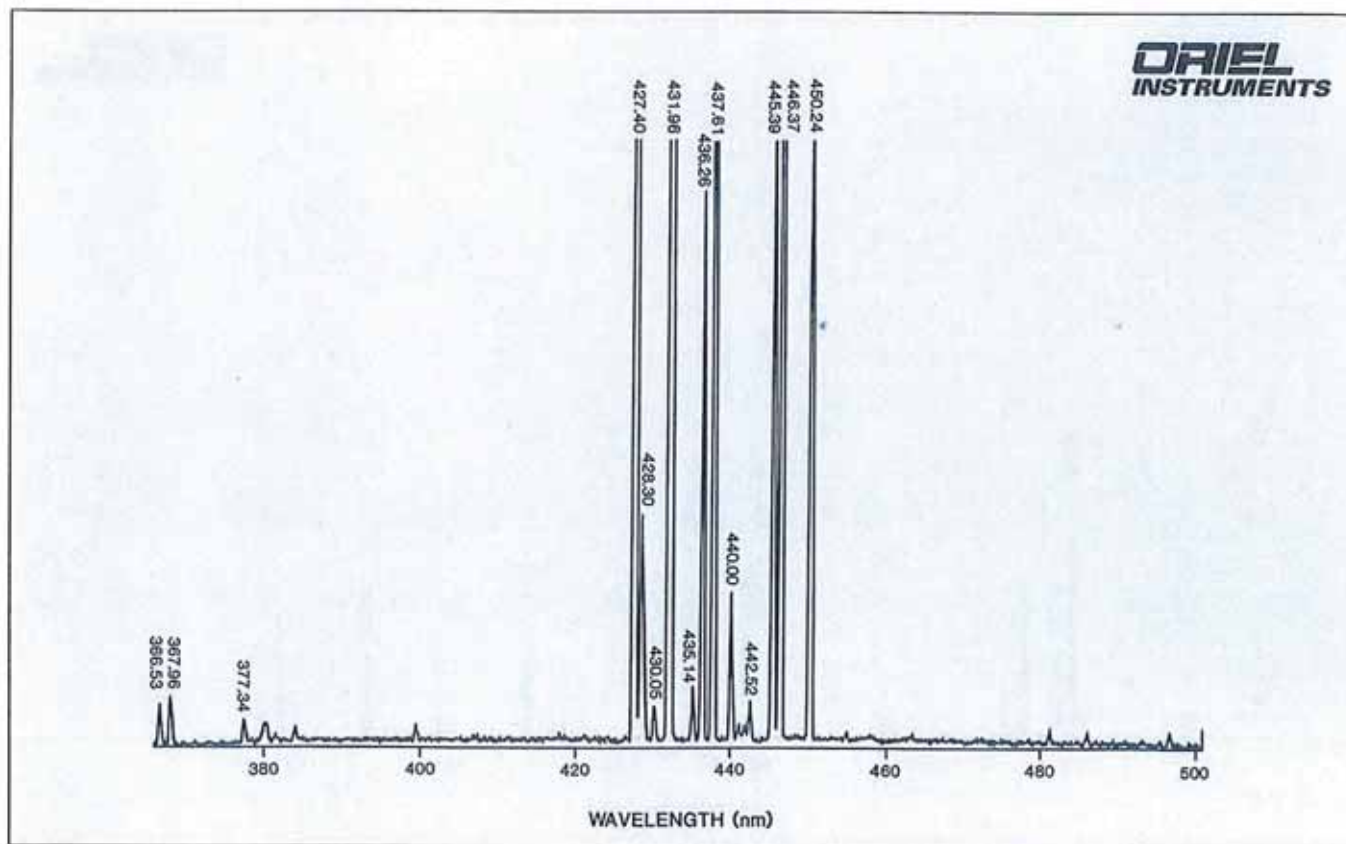


Fig. 6 Typical line output of 6031 Krypton Lamp. Relative intensities vary with operating conditions.

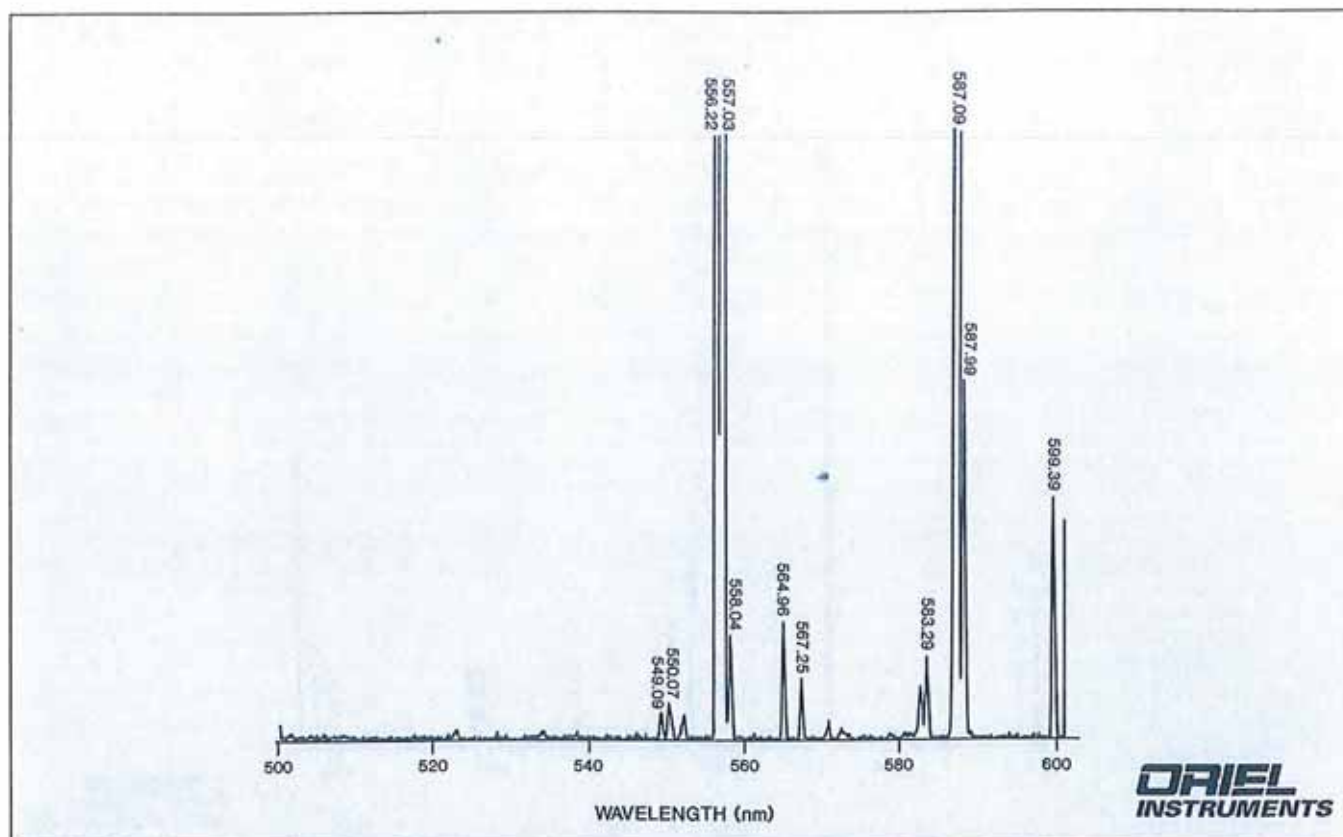


Fig. 7 Typical line output of 6031 Krypton Lamp. Relative intensities vary with operating conditions.

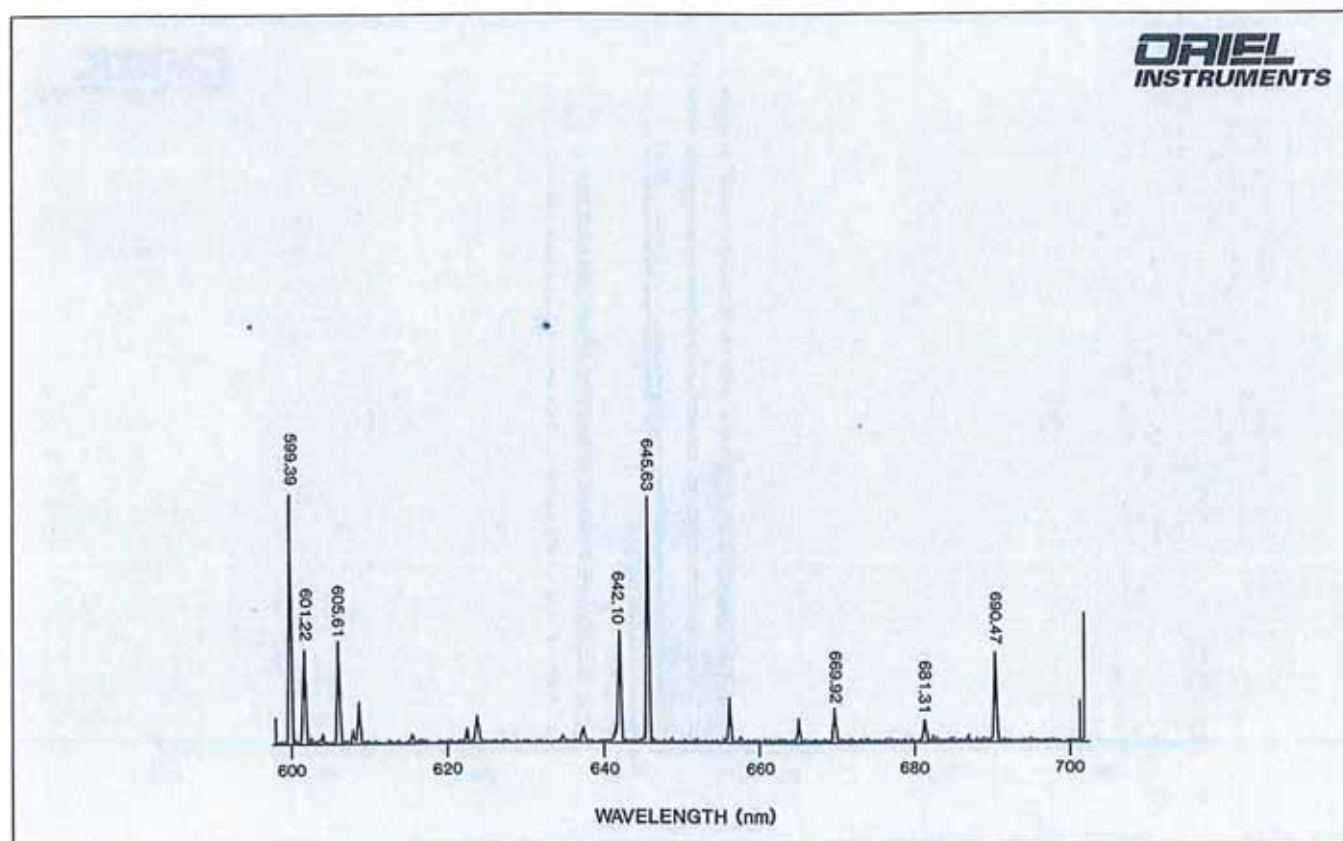


Fig. 8 Typical line output of 6031 Krypton Lamp. Relative intensities vary with operating conditions.

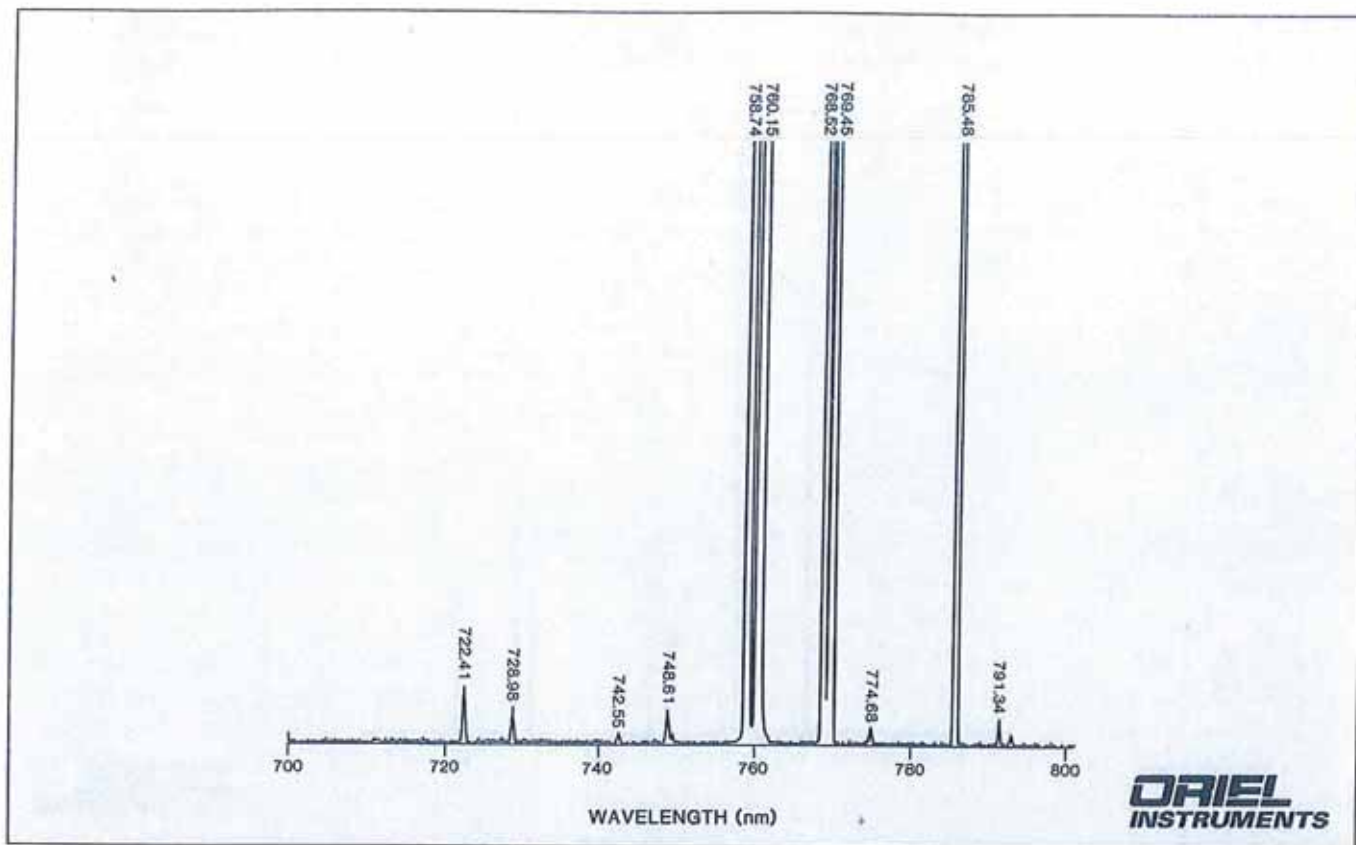


Fig. 9 Typical line output of 6031 Krypton Lamp. Relative intensities vary with operating conditions.

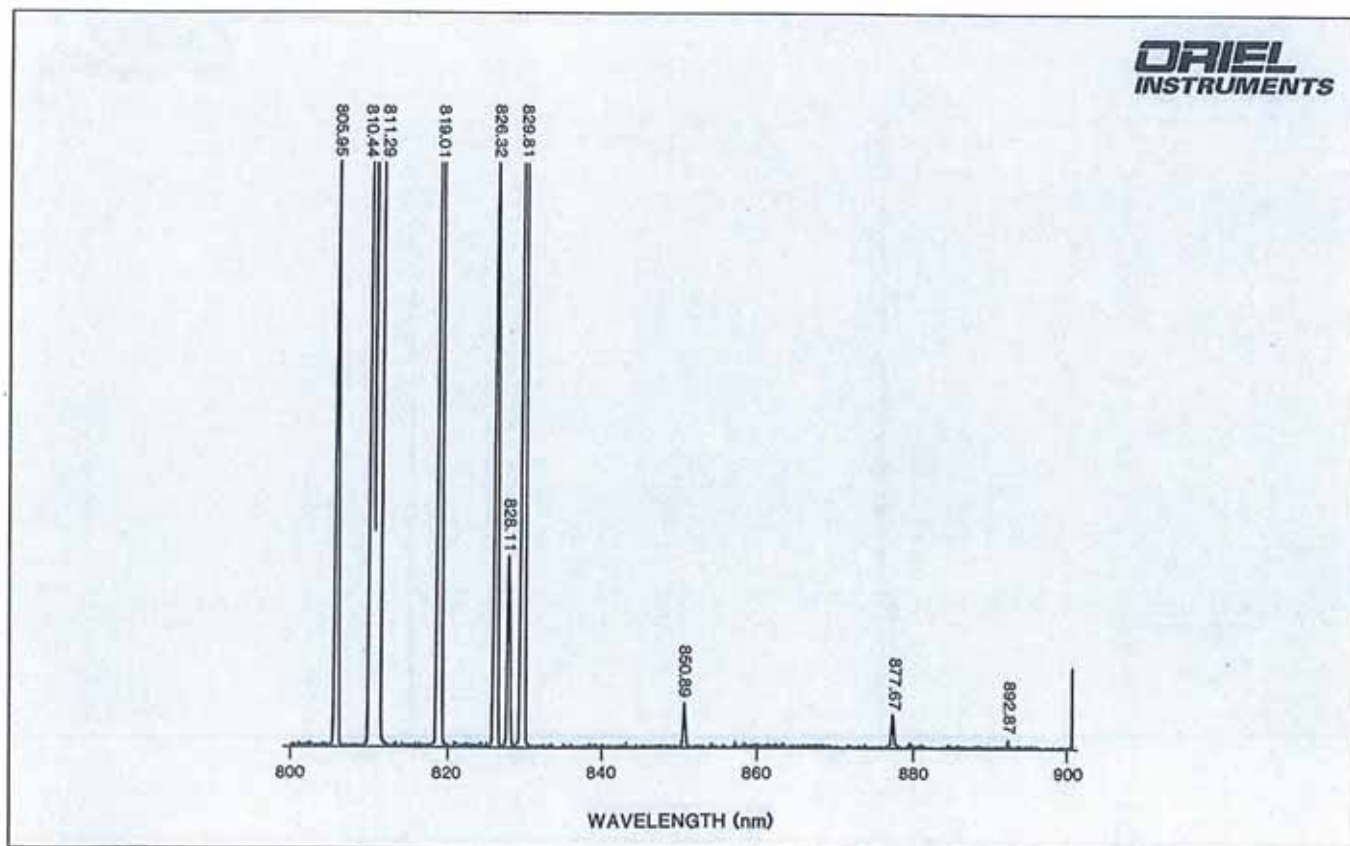


Fig. 10 Typical line output of 6031 Krypton Lamp. Relative intensities vary with operating conditions.

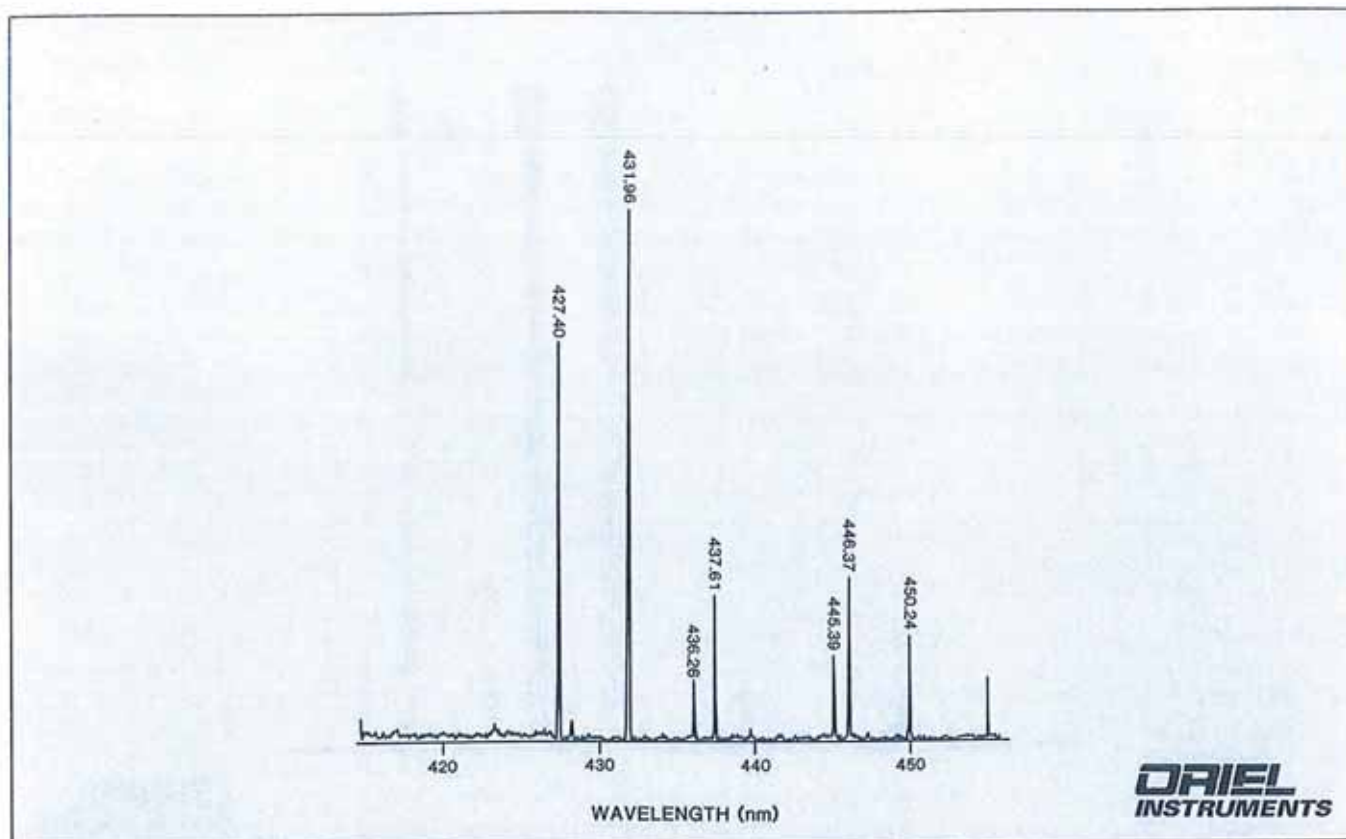


Fig. 11 Typical line output of 6031 Krypton Lamp. Relative intensities vary with operating conditions.

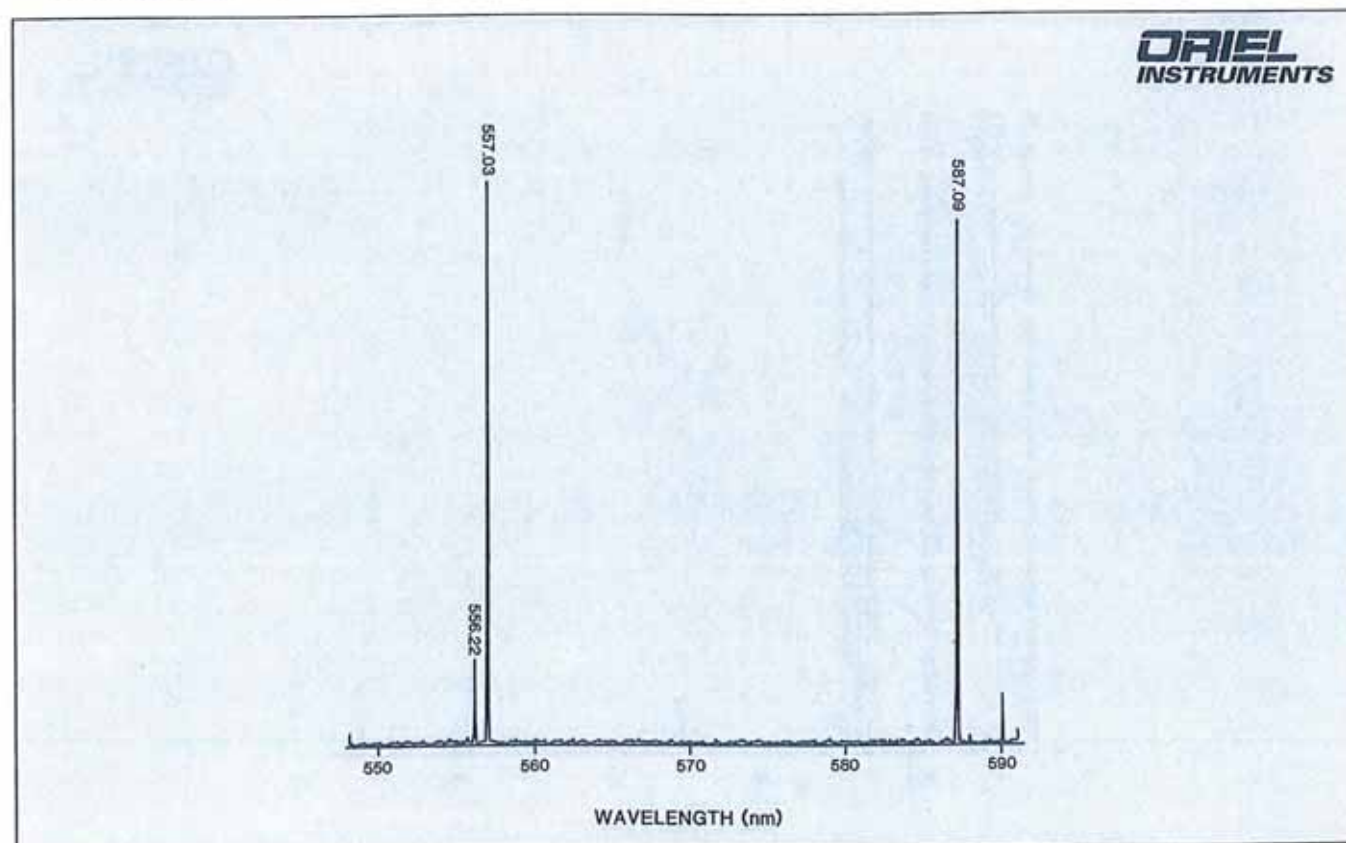


Fig. 12 Typical line output of 6031 Krypton Lamp. Relative intensities vary with operating conditions.

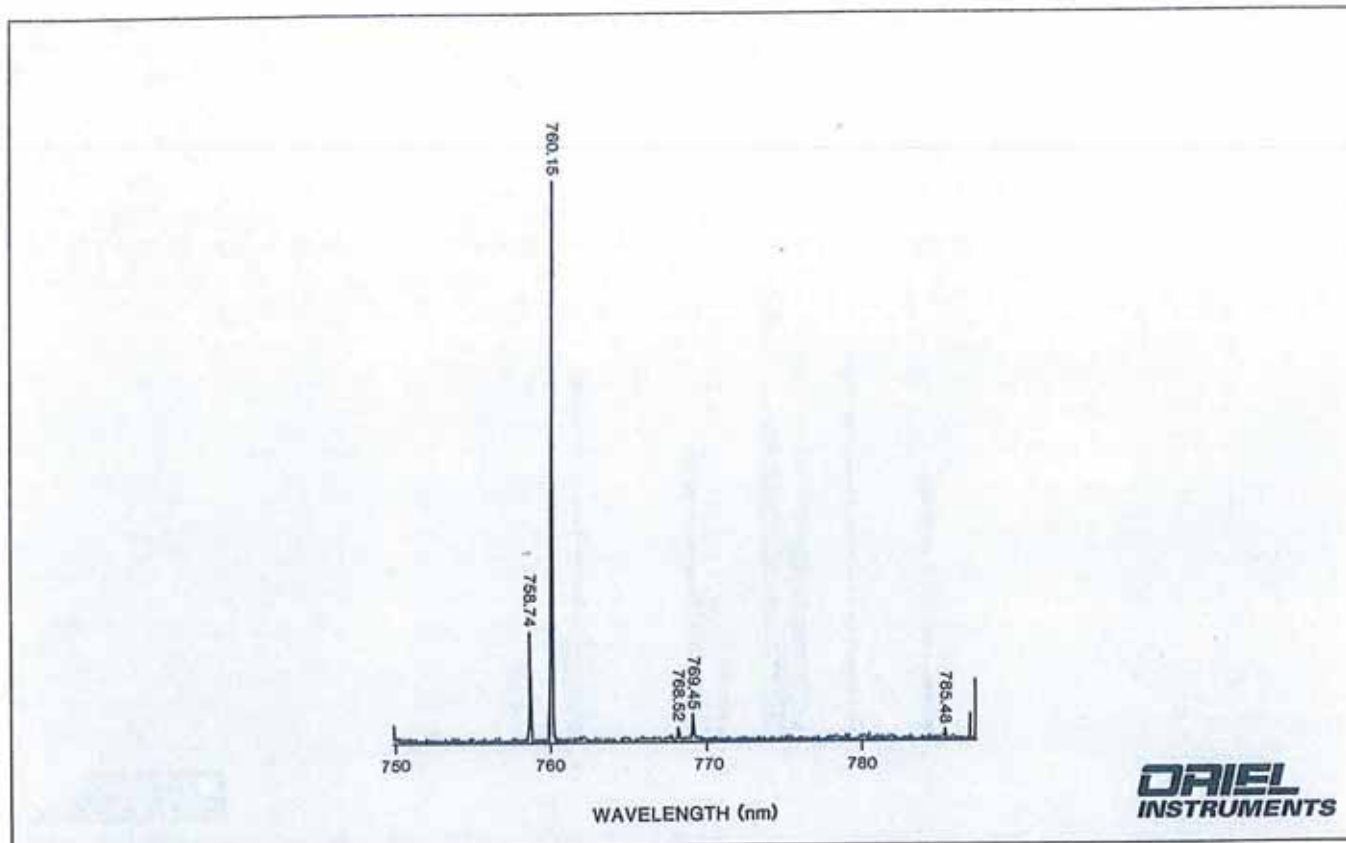


Fig. 13 Typical line output of 6031 Krypton Lamp. Relative intensities vary with operating conditions.

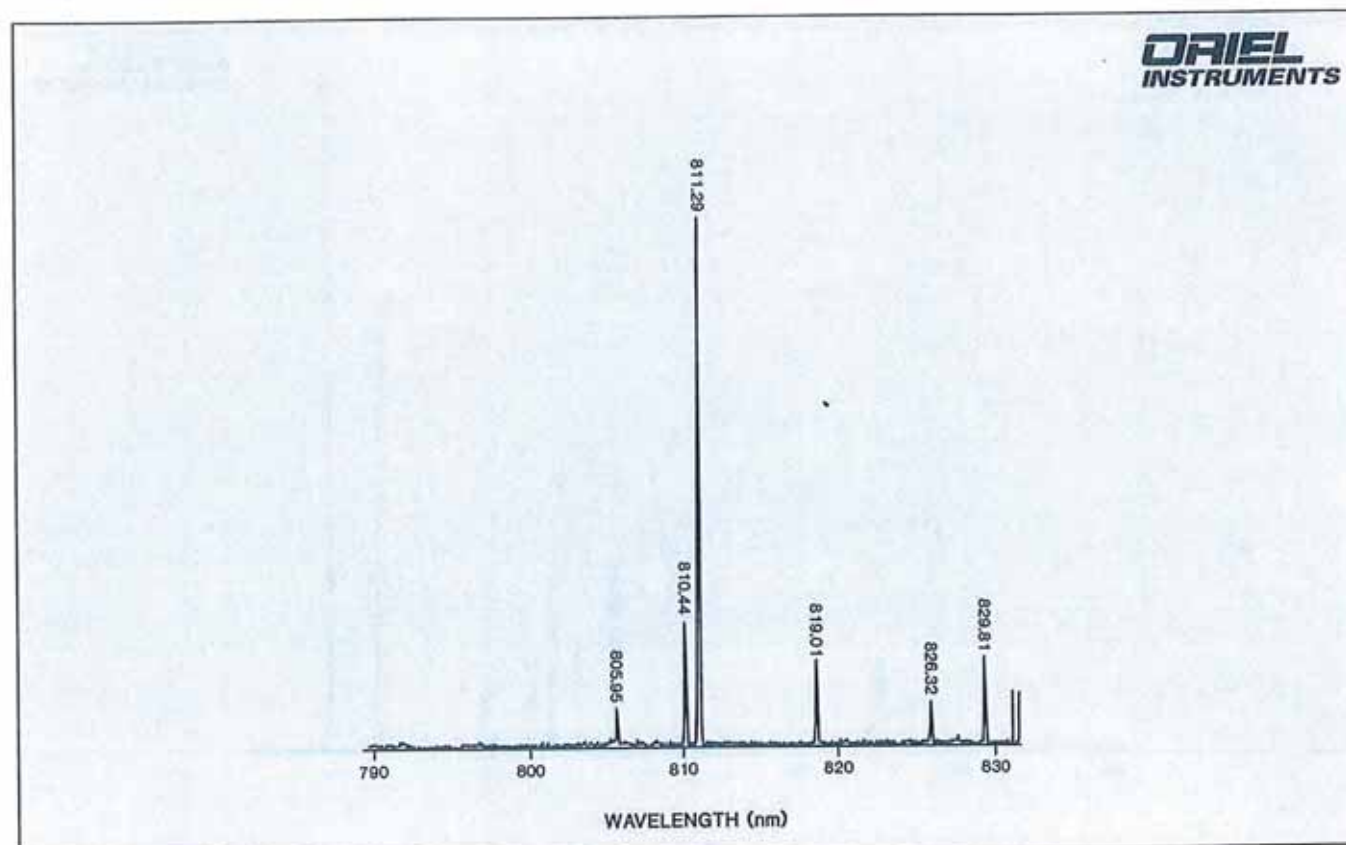


Fig. 14 Typical line output of 6031 Krypton Lamp. Relative intensities vary with operating conditions.

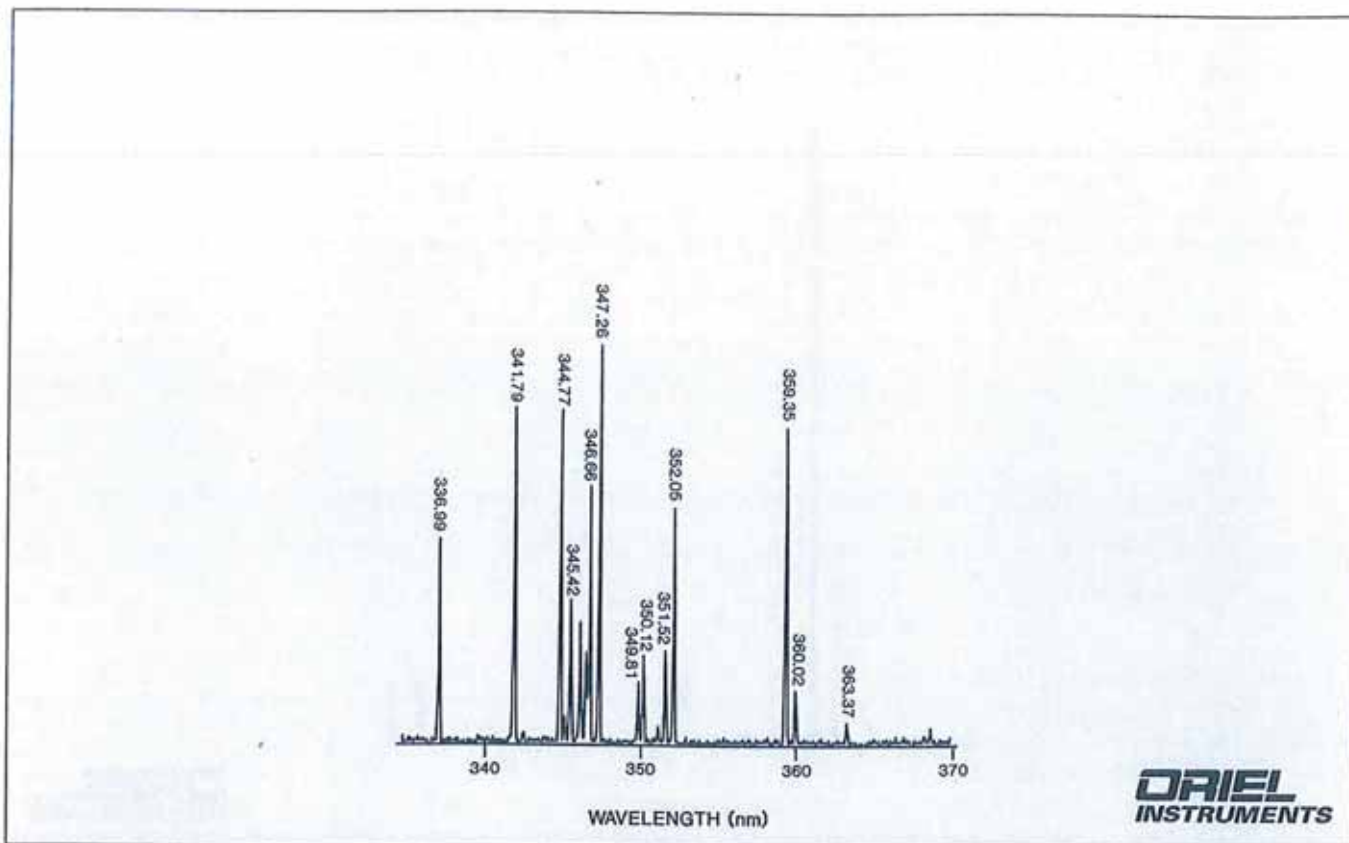


Fig. 15 Typical line output of 6032 Neon Lamp. Relative intensities vary with operating conditions.

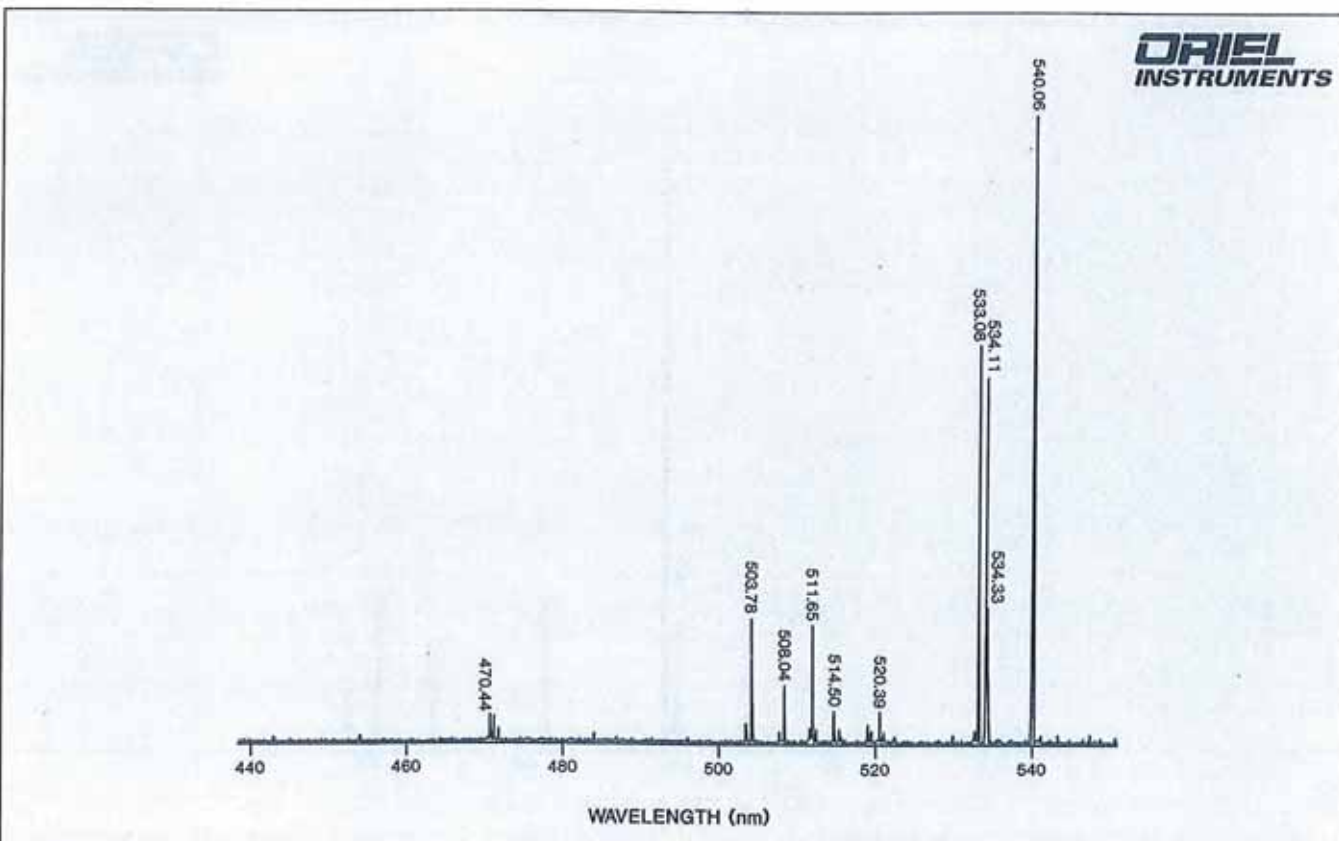


Fig. 16 Typical line output of 6032 Neon Lamp. Relative intensities vary with operating conditions.

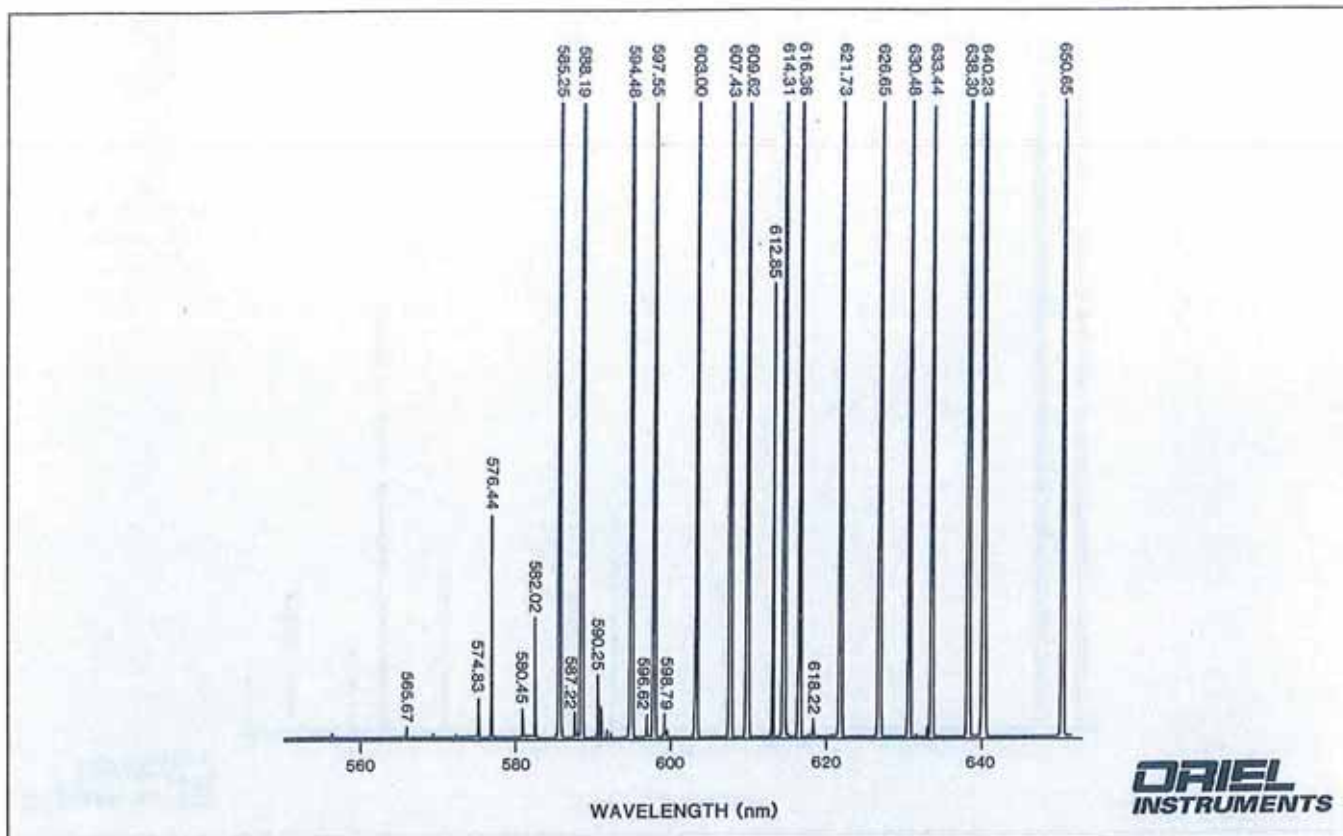


Fig. 17 Typical line output of 6032 Neon Lamp. Relative intensities vary with operating conditions.

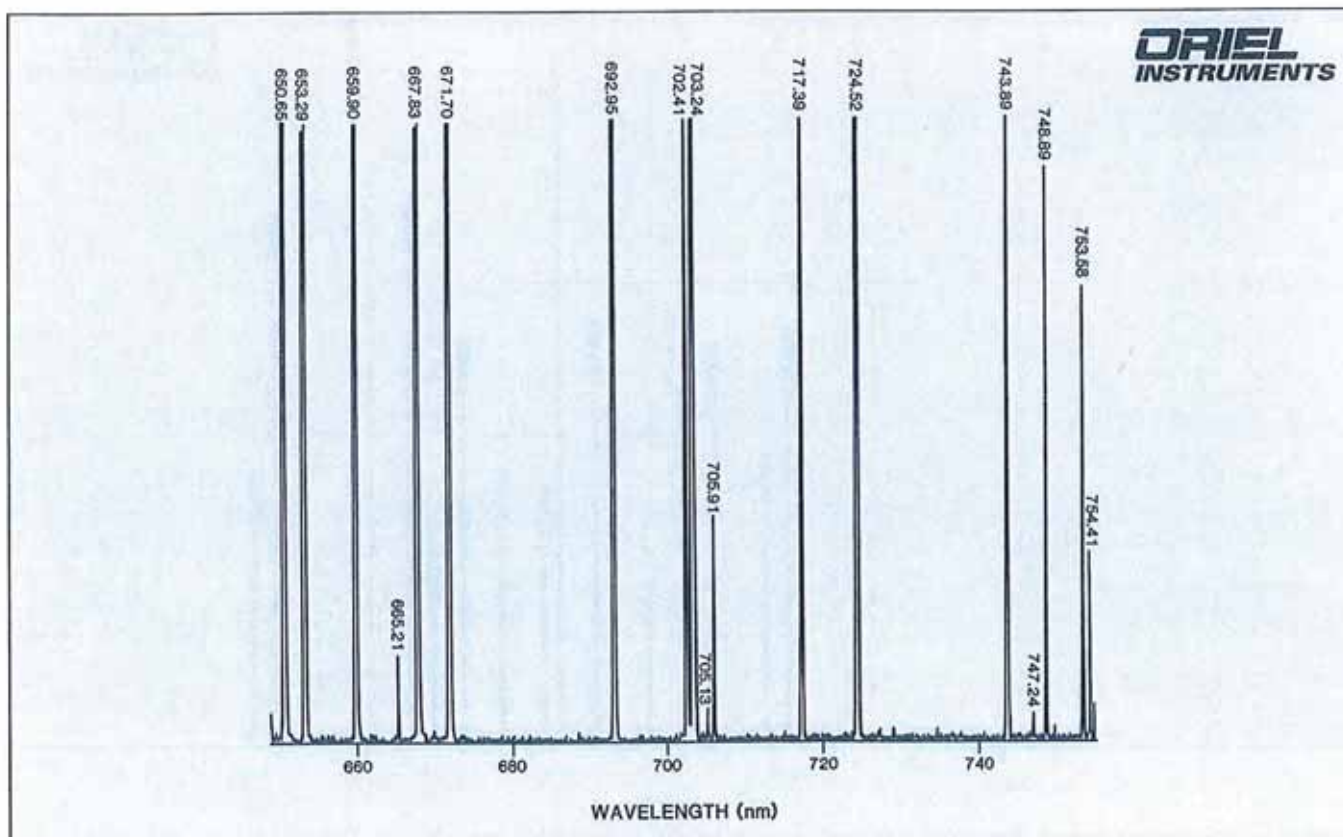


Fig. 18 Typical line output of 6032 Neon Lamp. Relative intensities vary with operating conditions.

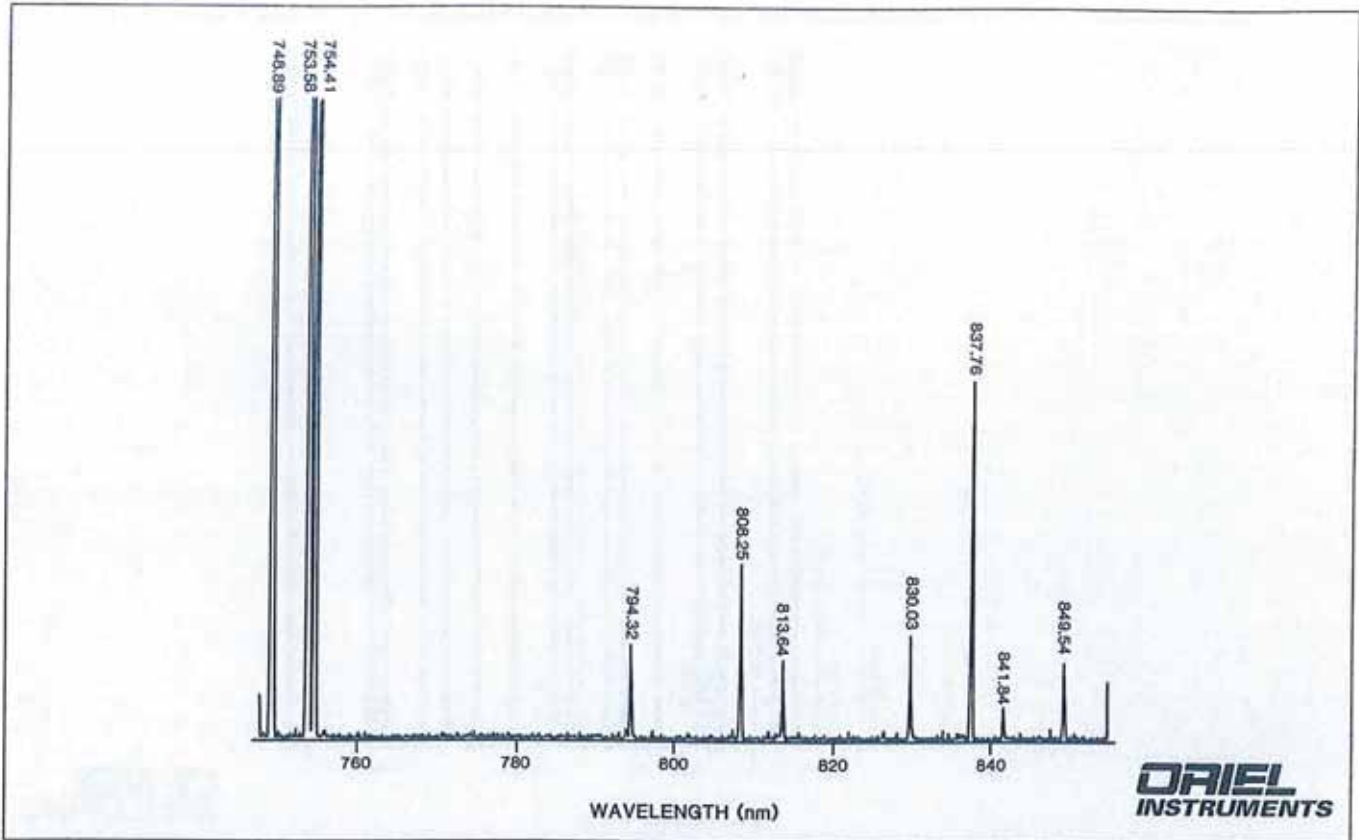


Fig. 19 Typical line output of 6032 Neon Lamp. Relative intensities vary with operating conditions.

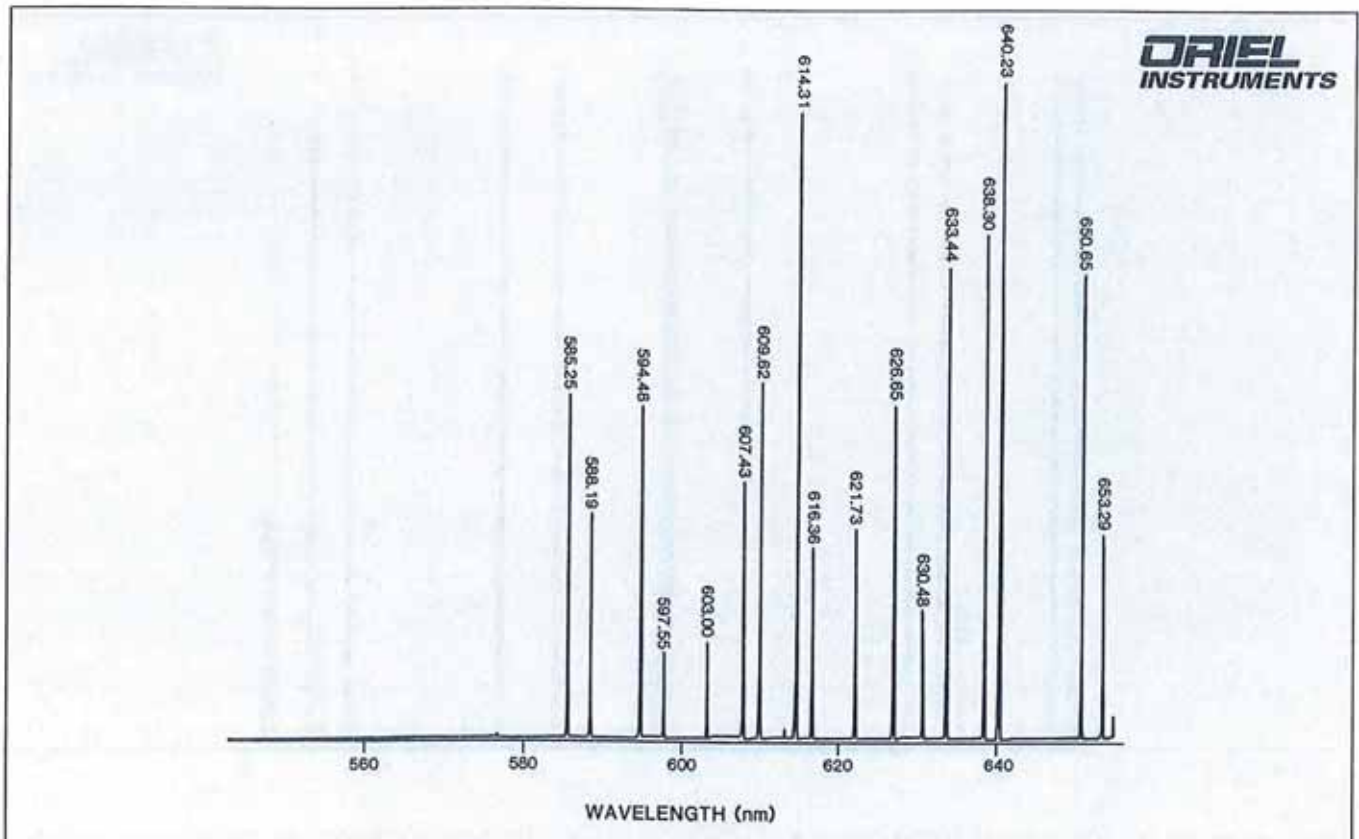


Fig. 20 Typical line output of 6032 Neon Lamp. Relative intensities vary with operating conditions.

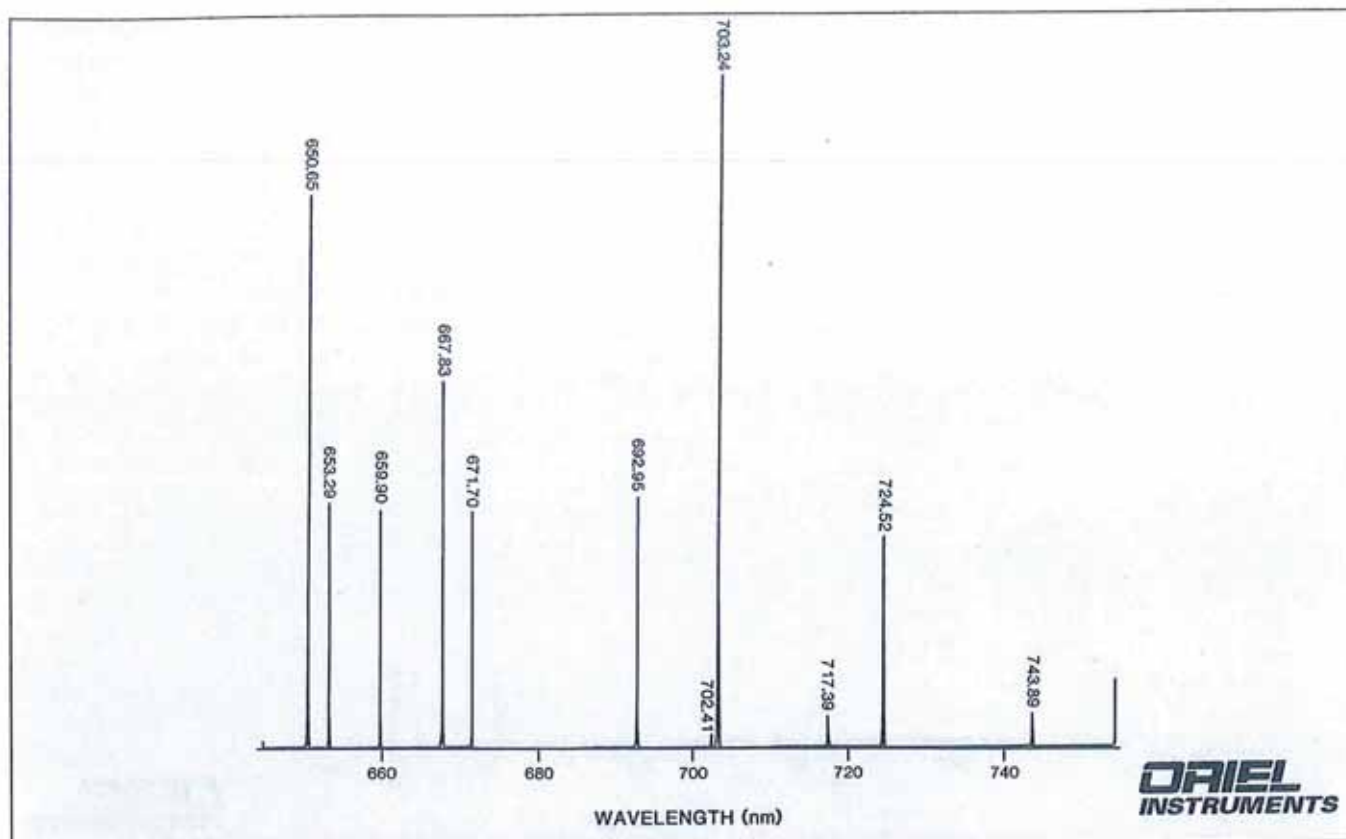


Fig. 21 Typical line output of 6032 Neon Lamp. Relative intensities vary with operating conditions.

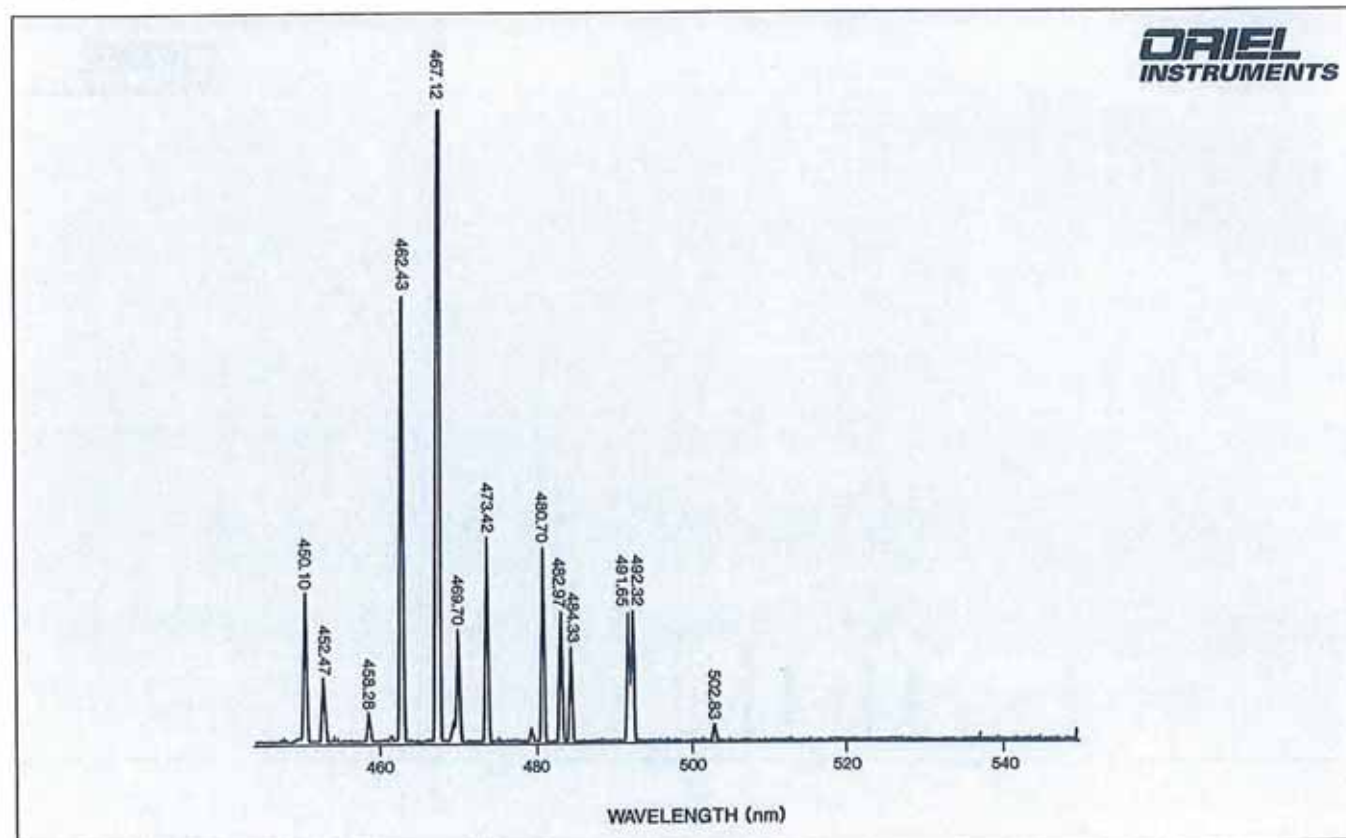


Fig. 22 Typical line output of 6033 Xenon Lamp. Relative intensities vary with operating conditions.

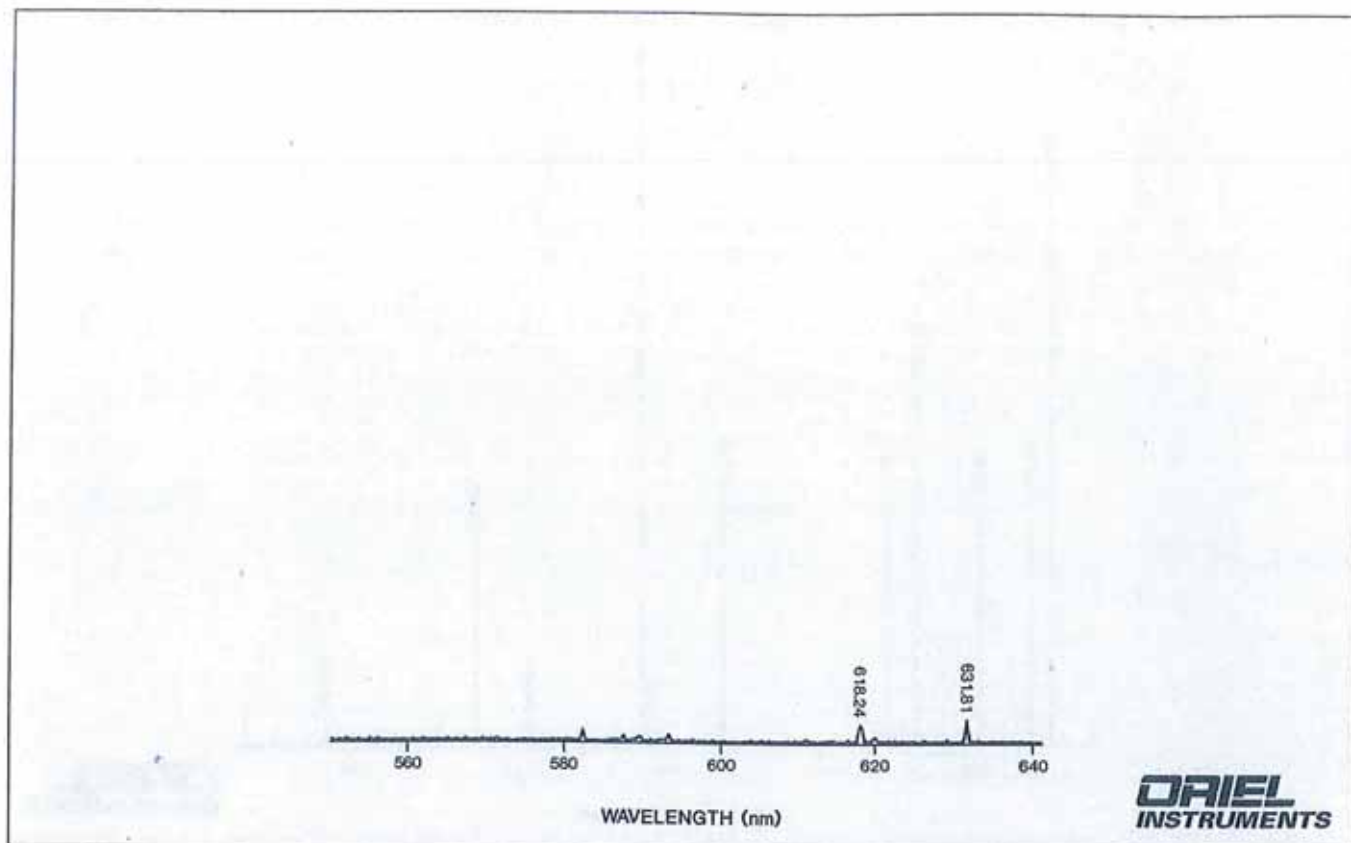


Fig. 23 Typical line output of 6033 Xenon Lamp. Relative intensities vary with operating conditions.

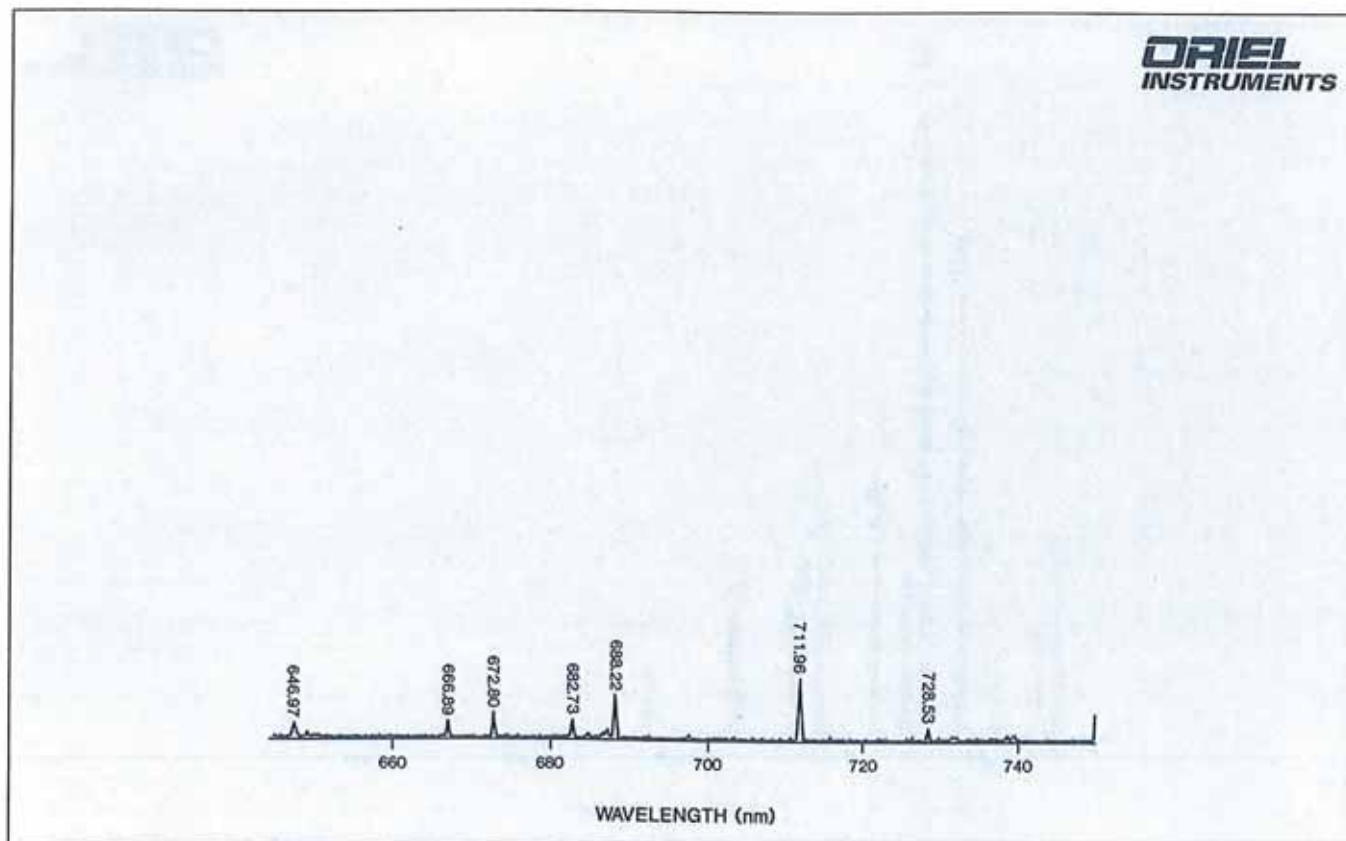


Fig. 24 Typical line output of 6033 Xenon Lamp. Relative intensities vary with operating conditions.

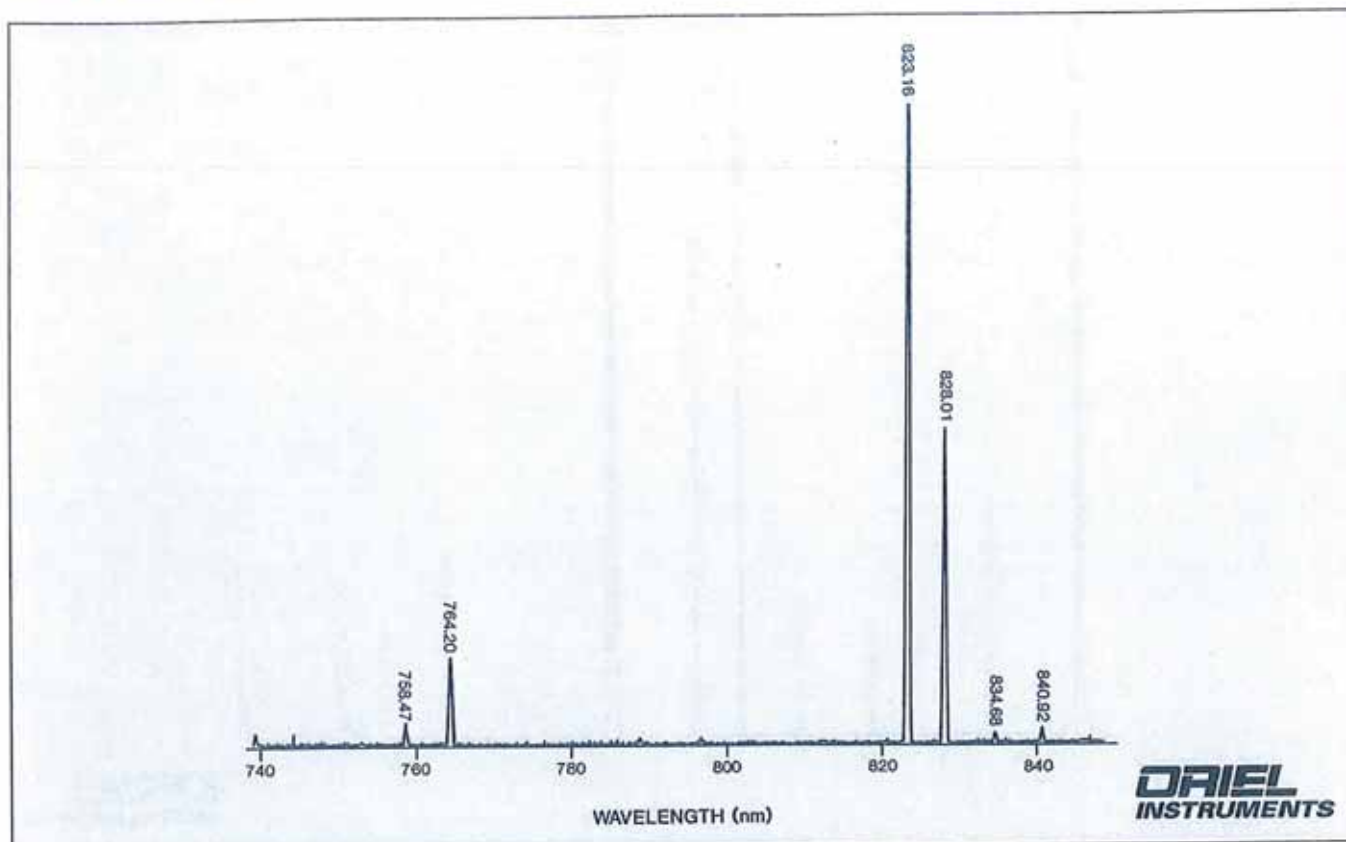


Fig. 25 Typical line output of 6033 Xenon Lamp. Relative intensities vary with operating conditions.

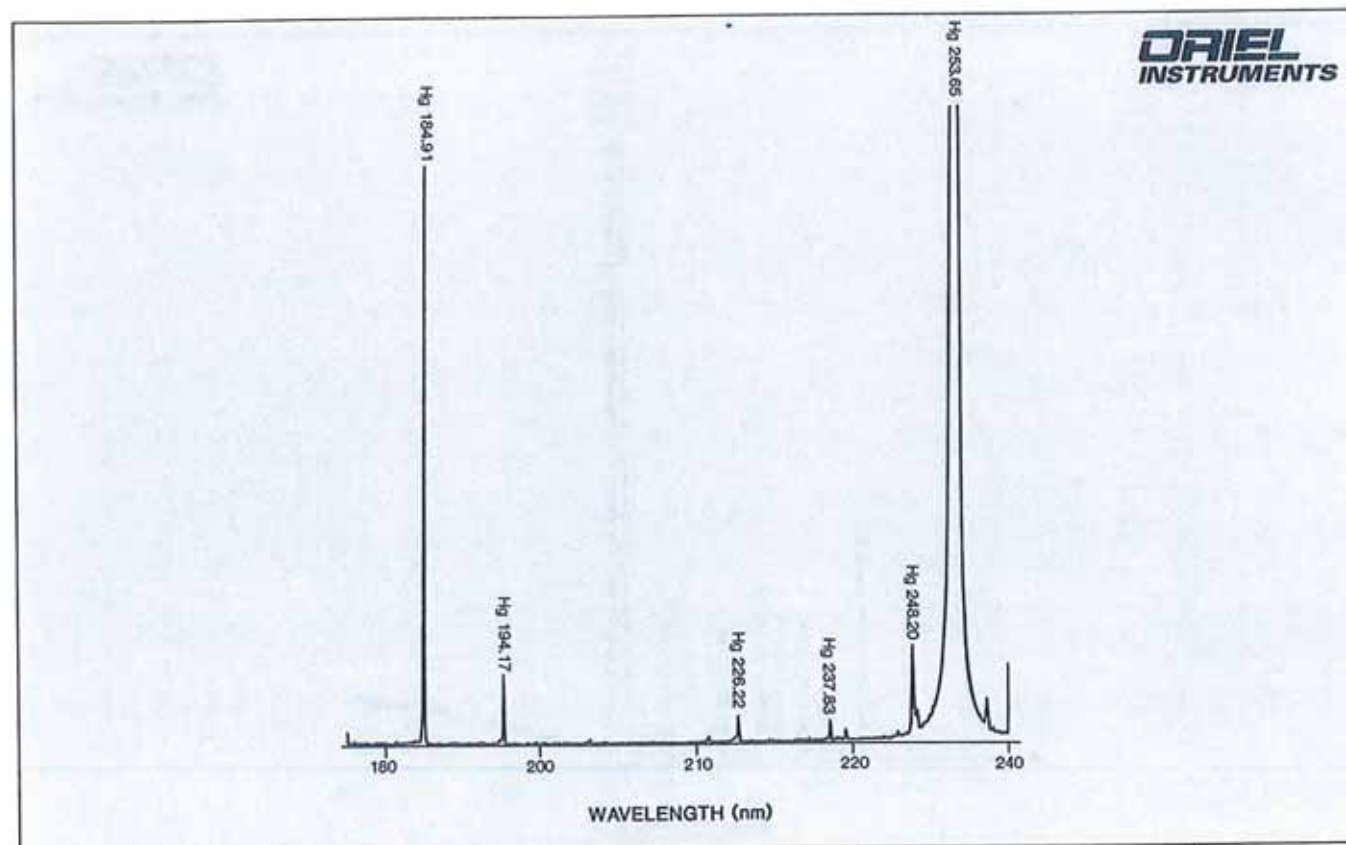


Fig. 26 Typical line output of 6034 Mercury-Neon and 6035 Mercury (Argon) Lamps. Relative intensities vary with operating conditions.

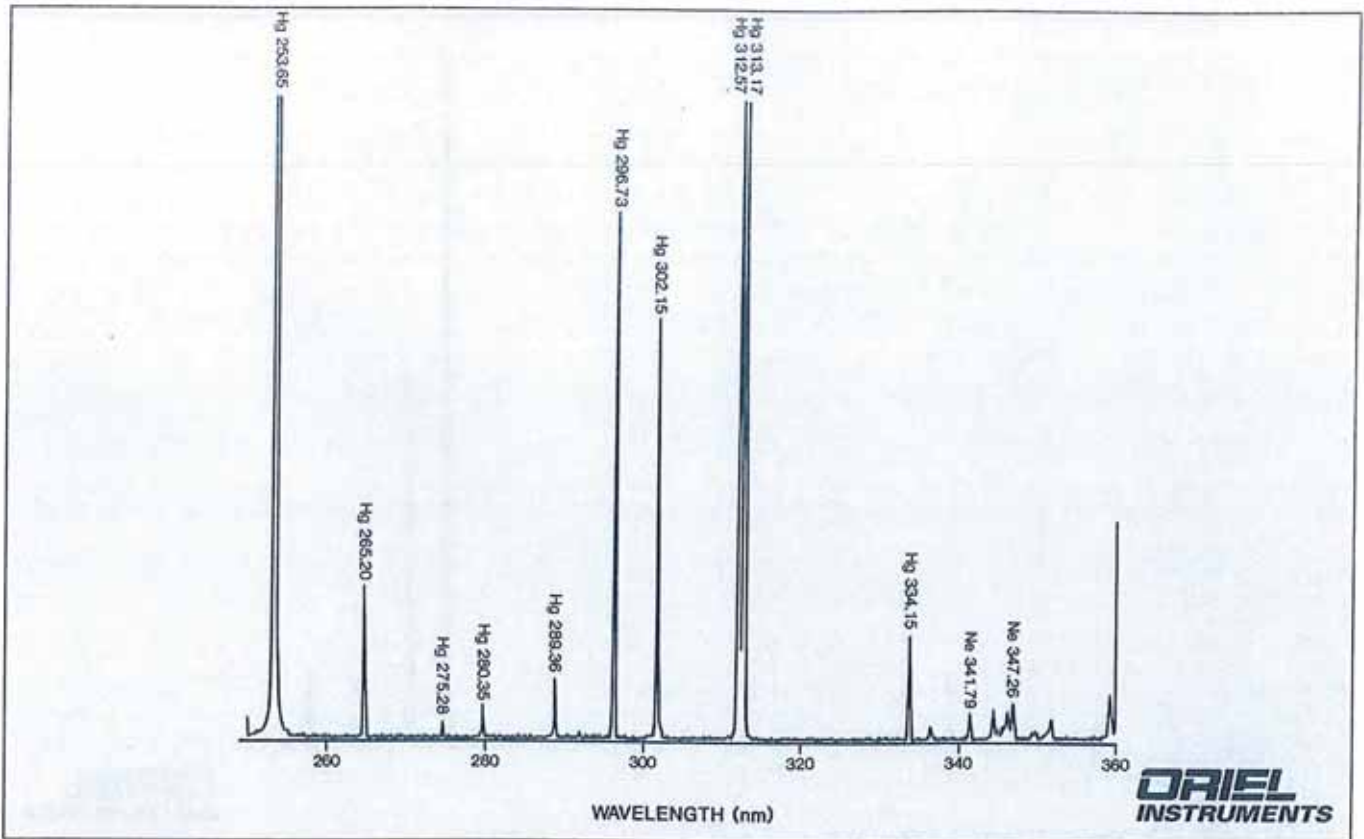


Fig. 27 Typical line output of 6034 Mercury-Neon Lamp. (Forced air cooling was used to bring out the Ne lines.) The Hg lines are present in the output of the 6035 Mercury (Argon) Lamp. Relative intensities vary with operating conditions.

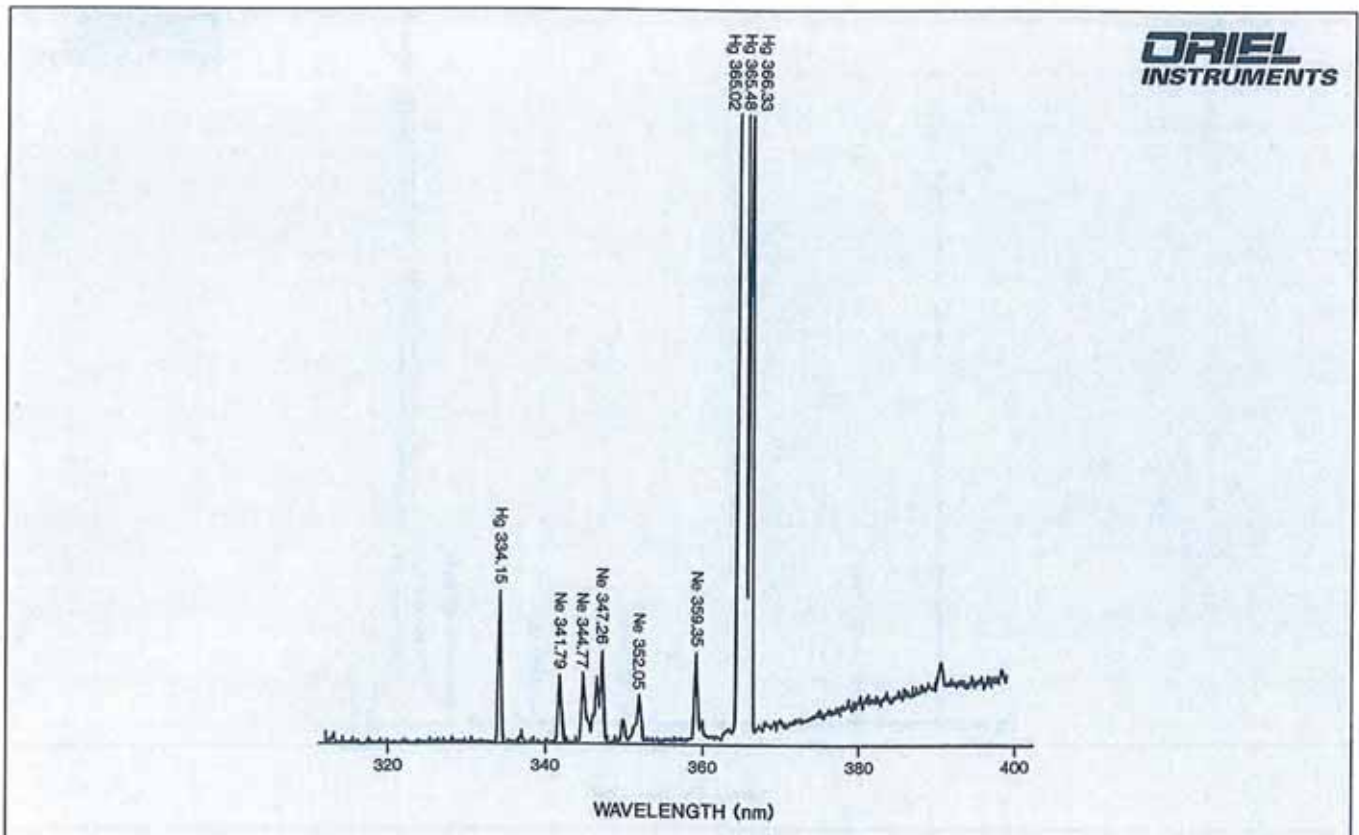


Fig. 28 Typical line output of 6034 Mercury-Neon Lamp. (Forced air cooling was used to bring out the Ne lines.) The Hg lines are present in the output of the 6035 Mercury (Argon) Lamp. Relative intensities vary with operating conditions.

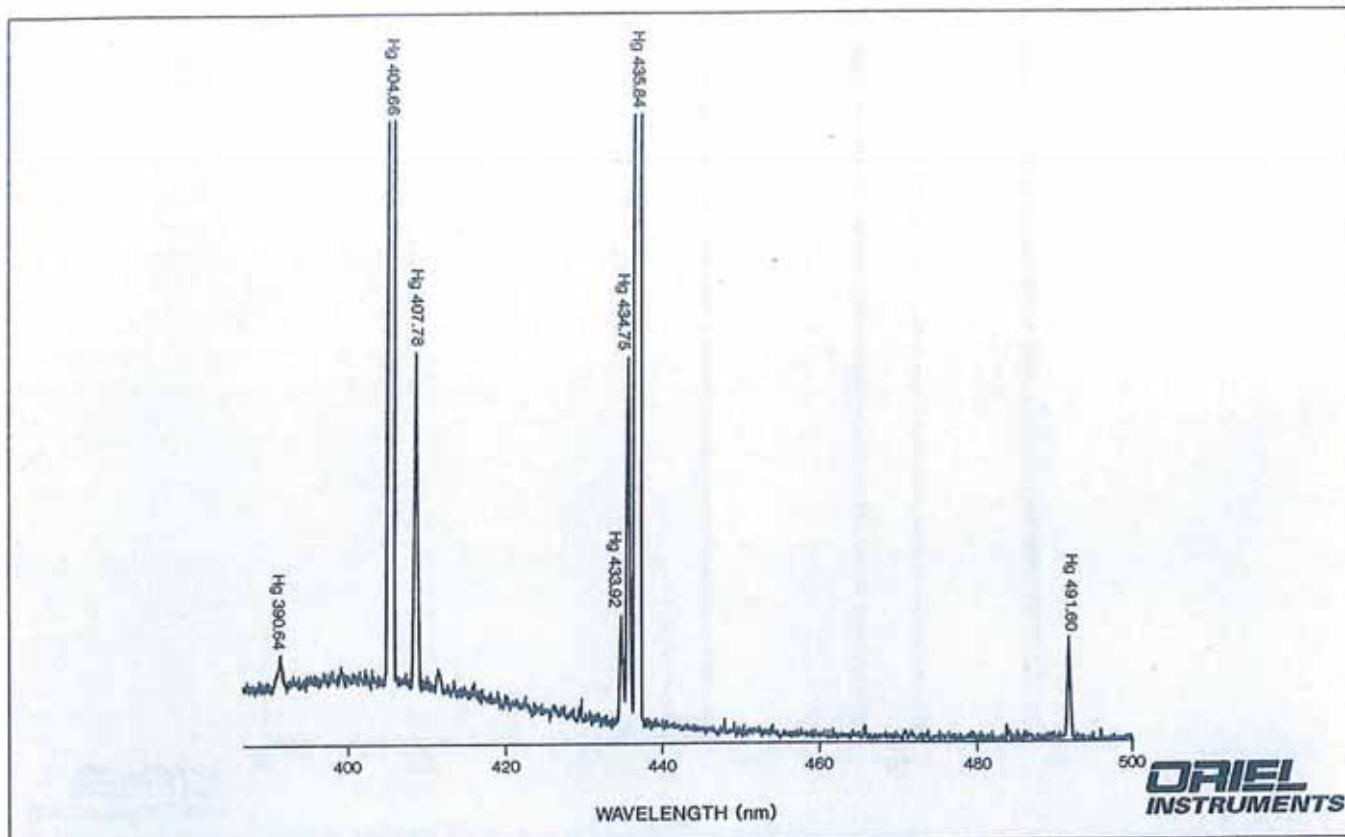


Fig. 29 Typical line output of 6034 Mercury-Neon and 6035 Mercury (Argon) Lamps. Relative intensities vary with operating conditions.

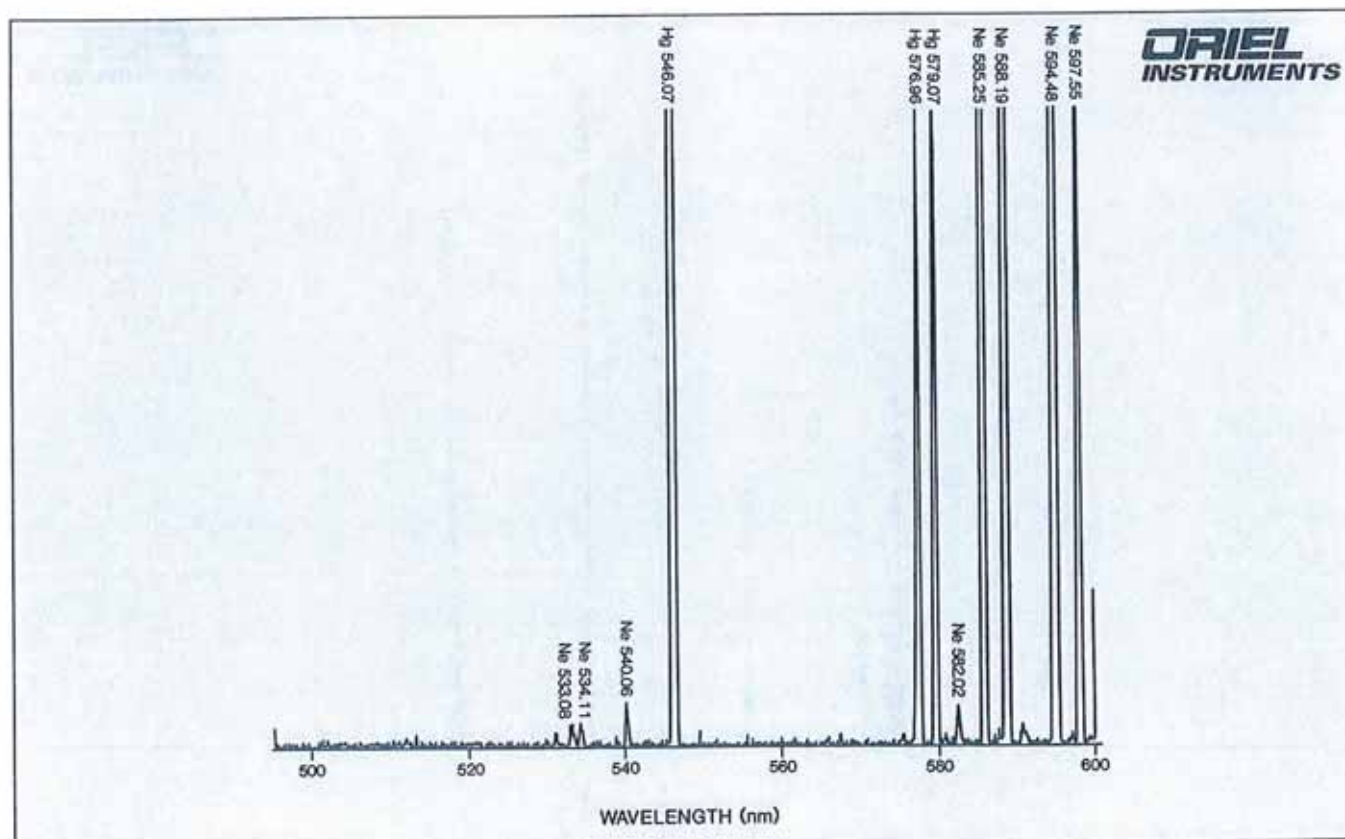


Fig. 30 Typical line output of 6034 Mercury-Neon Lamp. (Forced air cooling was used to bring out the Ne lines.) The Hg lines are present in the output of the 6035 Mercury (Argon) Lamp. Relative intensities vary with operating conditions.

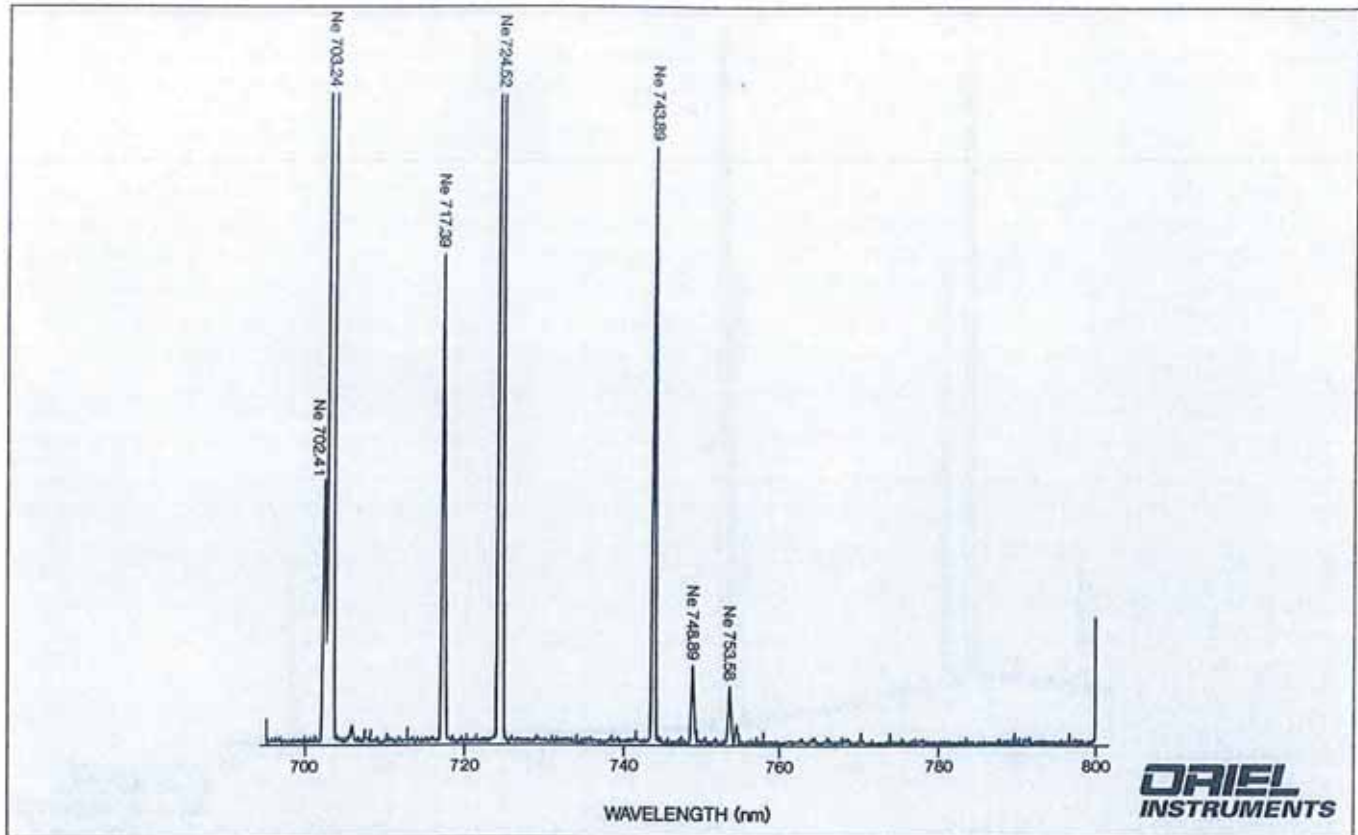


Fig. 31 Typical line output of 6034 Mercury-Neon Lamp. (Forced air cooling was used to bring out the Ne lines.) Relative intensities vary with operating conditions.

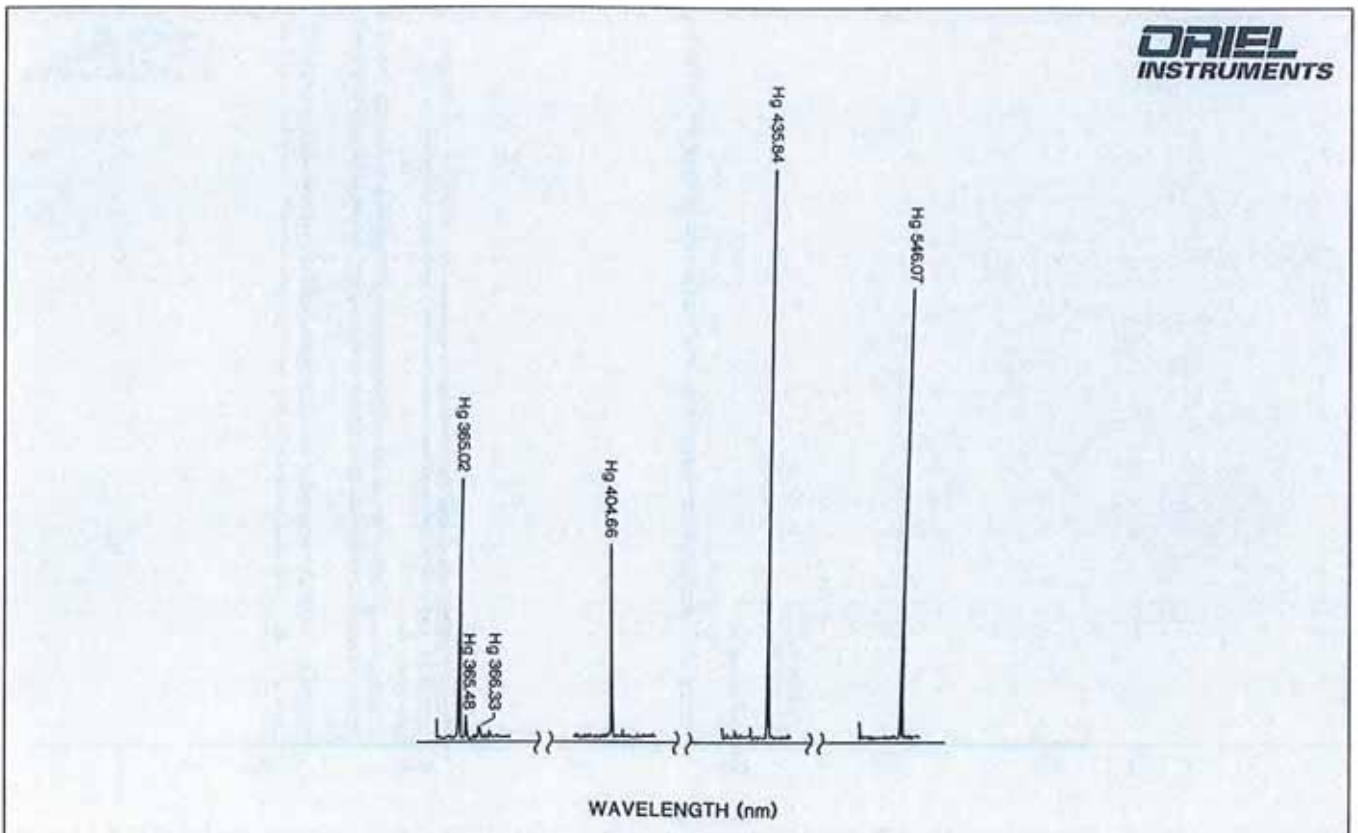


Fig. 32 Typical line output of 6034 Mercury-Neon and 6035 Mercury (Argon) Lamps. (Forced air cooling was used to bring out the Ne lines.) Relative intensities vary with operating conditions.

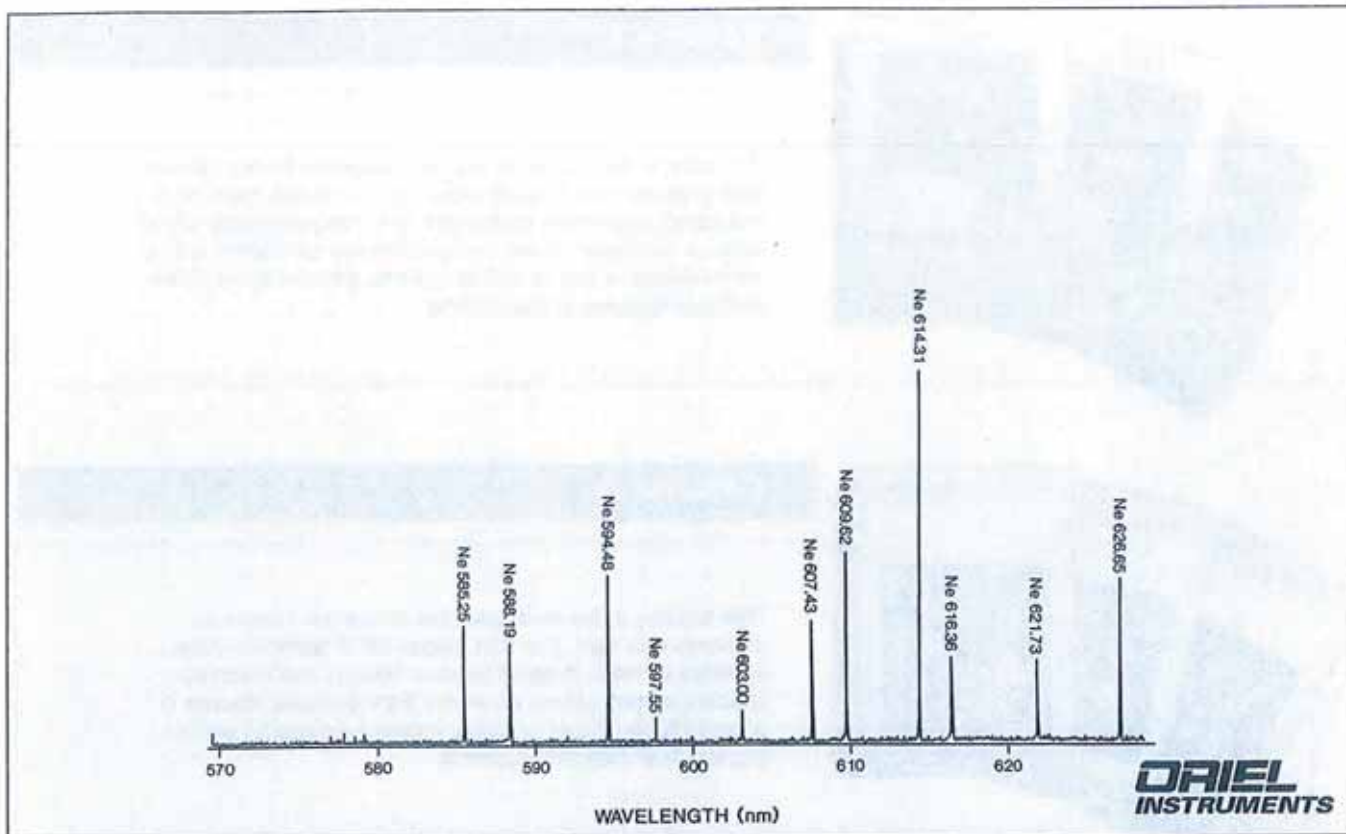


Fig. 33 Typical line output of 6034 Mercury-Neon Lamp. (Forced air cooling was used to bring out the Ne lines.) Relative intensities vary with operating conditions.

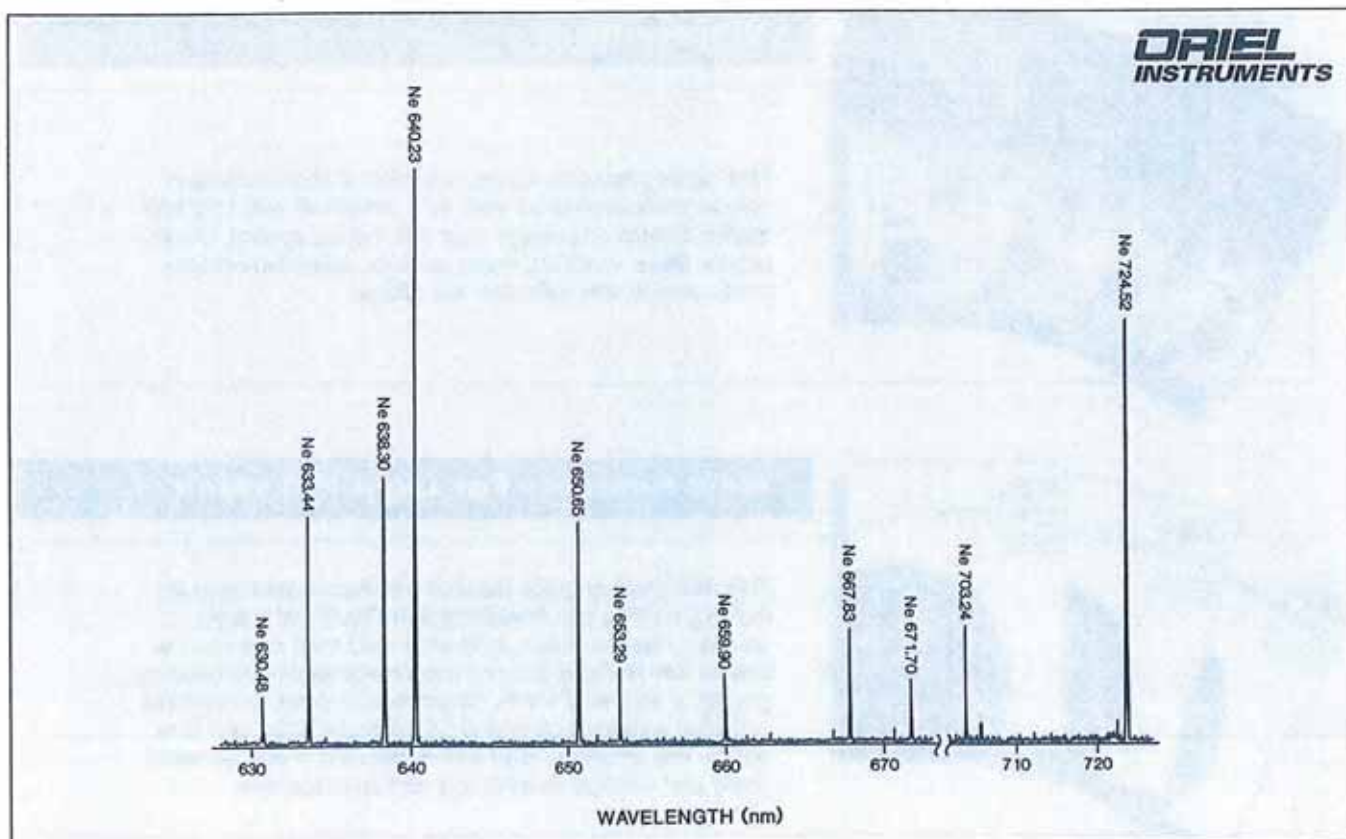


Fig. 34 Typical line output of 6034 Mercury-Neon Lamp. (Forced air cooling was used to bring out the Ne lines.) Relative intensities vary with operating conditions.